

FORGE: REACTION-GUIDED GENERATIVE DESIGN OF IONIZABLE LIPIDS

Anonymous authors

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ABSTRACT

Ionizable lipids are central components of lipid nanoparticles for RNA delivery. Their discovery often relies on synthesizing and screening libraries assembled from predefined reactants, which restricts the precursor structures available for design. Whole-molecule generation can extend this search, but verifying assembly requires precursor assignments and reaction information. We introduce FORGE (Flow-matched, Open-ended, Route-resolved Generation and Exploration), which uses known reaction chemistry to design ionizable lipids without restricting generation to a fixed reactant catalogue. Each reaction program specifies precursor roles, reaction-core positions, transformation order and coarse molecular architecture. FORGE generates the complete molecule at atom-and-bond resolution without component identifiers, stored precursor graphs or fragment tokens. Reverse decomposition and exact forward replay verify the implied assembly. We evaluated one shared model on an AGILE-type amine–aldehyde–isocyanide Ugi 3-CR, repeated azo-Michael addition and repeated reductive amination. Across three independent seeds, verified exact level-one (exact-L1) yields were $96.4 \pm 2.1\%$, $72.2 \pm 1.6\%$ and $52.3 \pm 3.4\%$, versus $34.8 \pm 7.1\%$, $19.2 \pm 4.1\%$ and $37.0 \pm 2.2\%$ for matched whole-molecule generation followed by post-hoc verification. FORGE also generated exact-L1 products with precursor identities absent from the corresponding training component catalogues across all three reactions. These results establish reaction-program conditioning as a means of generating complete lipid graphs with verified assembly and recovered precursors beyond the training inventory.

1 INTRODUCTION

Lipid nanoparticles (LNPs) deliver nucleic-acid therapeutics including small interfering RNA, mRNA vaccines, and *in vivo* genome-editing reagents (Adams et al., 2018; Polack et al., 2020; Baden et al., 2021; Gillmore et al., 2021). Within each LNP, the ionizable lipid supports nucleic-acid complexation, particle assembly, and endosomal escape, making its chemical structure a major determinant of delivery (Cullis & Hope, 2017; Hou et al., 2021). Lipid structure and stereochemistry can alter delivery (Han et al., 2021; Hatit et al., 2023), and formulation composition can change tissue targeting (Cheng et al., 2020). Combinatorial discovery therefore combines modular precursor libraries with synthesis and experimental screening of the resulting LNPs (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). This strategy supports systematic synthesis and testing within defined chemical families. The precursor inventory sets the molecular structures available to that search.

De novo molecular generation can expand the design space beyond a screened library. Synthesis-aware methods including SynNet, RGFN and SynFlowNet construct molecules by selecting reactants and applying predefined transformations (Gao et al., 2022; Koziarski et al., 2024; Cretu et al., 2025). These methods retain an explicit assembly route and restrict precursor selection to a supplied inventory. SynCoGen jointly generates building-block graphs, reaction assignments and atomic coordinates (Rekesh et al., 2026). Lipid-specific synthesis-DAG and tree-search methods also generate products from available building blocks (Ou et al., 2024; Zhao et al., 2024). Whole-molecule models such as DeFoG and GenMol generate structures without a prescribed precursor inventory (Qin et al., 2025; Lee et al., 2025); their molecular outputs require a separate assessment of assembly and synthesizability (Gao & Coley, 2020). The capability we seek is complete molecular generation under specified assembly chemistry with precursor identities left open.

Here, we introduce **FORGE** (**F**low-matched, **O**pen-ended, **R**oute-resolved **G**eneration and **E**xploration), a generative model that uses known reaction chemistry to design ionizable lipids beyond a fixed reactant catalogue. Each reaction program contains the assembly chemistry, precursor roles, reaction-core positions, transformation order, and coarse molecular architecture while leaving precursor identities unspecified. **FORGE** generates complete molecular graphs at atom-and-bond resolution without component identifiers, stored precursor graphs, or fragment tokens. We then decompose each completed graph into its implied precursors and require exact forward replay of the specified assembly reaction to reconstruct the generated molecule. For unavailable precursors, a bounded recursive search returns either a computational synthesis dossier supported by reaction and supplier evidence or an explicit unresolved or search-censored outcome.

We evaluated one shared **FORGE** model across an **AGILE**-type amine–aldehyde–isocyanide Ugi 3-CR, repeated aza-Michael addition, and repeated reductive amination (Xu et al., 2024; Li et al., 2023; Xue et al., 2024). Across three independent seeds, **FORGE** achieved verified exact-L1 yields of $96.4 \pm 2.1\%$, $72.2 \pm 1.6\%$, and $52.3 \pm 3.4\%$, respectively, exceeding matched whole-molecule generation followed by post-hoc verification in every program. **FORGE** also produced verified exact assemblies with recovered precursor constitutions absent from the corresponding training catalogues. These experiments tested generation across three specified reaction programs; zero-shot generation under unseen assembly chemistry remains untested.

The paper makes three contributions:

- **FORGE** generates complete molecular graphs using known reaction chemistry while leaving precursor identities open and does not select from stored reactants or fragment vocabularies.
- Matched controls across three assembly programs showed that reaction-program conditioning substantially increased exact transform-consistent generation relative to whole-molecule generation followed by post-hoc verification.
- Across all three programs, **FORGE** generated verified exact assemblies with recovered precursor constitutions absent from the corresponding training component catalogues. This establishes component-level expansion beyond the fixed training inventories; strong product-level catalogue escape remains unclaimed.

2 PRELIMINARIES AND PROBLEM FORMULATION

2.1 IONIZABLE LIPIDS AND MOLECULAR ARCHITECTURE

Ionizable lipids (ILs) are amphiphilic molecules that combine protonatable groups with hydrophobic regions. Their architecture comprises an amine-containing headgroup, connecting linkers and one or more hydrophobic tails (Semple et al., 2010; Jayaraman et al., 2012). The headgroup’s charge depends on pH. Protonation under acidic formulation conditions supports association with anionic nucleic acids, and protonation in endosomes can contribute to membrane interactions involved in cargo release (Xu et al., 2024). The tails contribute hydrophobic association and packing. Linkers connect these ionizable and hydrophobic regions within the complete molecule.

An IL is one constituent of an LNP formulation, commonly combined with helper lipids, cholesterol and PEGylated lipids (Xu et al., 2024; Xue et al., 2024). Molecular structure and formulated-particle performance are therefore different levels of description. This study evaluates generated lipid constitutions and their assembly chemistry. Delivery activity requires evidence beyond molecular structure and exact reaction replay.

2.2 REACTION FAMILIES AND MODULAR LIPID ASSEMBLY

Combinatorial lipid synthesis varies precursor structures within a shared assembly chemistry, allowing a library to explore different headgroups, linkers and tails while retaining a common reaction pattern (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). A reaction family describes this bond-changing pattern. The three families used here differ both in the reactive groups they connect and in how precursor contributions are combined (Table 1; assembly schemes in Appendix D.1).

The Ugi variant is specifically the **AGILE**-type three-component reaction, with no carboxylic-acid reactant component; the reported procedure uses an acidic phosphorus catalyst. In the qualified

Table 1: **Assembly chemistry underlying the three reaction programs.** The named roles specify reactive functionality without selecting a precursor identity. Repeated programs encode the successive additions admitted by the corresponding registered transform.

Reaction family	Role-assigned precursors	Assembly pattern
AGILE-type Ugi 3-CR	Amine, aldehyde and isocyanide	Forms an α -amino amide combining three precursor contributions (Xu et al., 2024).
Repeated aza-Michael addition	Amine head and acrylate-bearing hydrophobic components	Amine addition to acrylate groups forms C–N bonds; successive additions attach hydrophobic components at available amine sites (Li et al., 2023).
Repeated reductive amination	Amine head and aldehyde-bearing hydrophobic components	Condensation followed by reduction forms C–N linkages, attaching hydrophobic components through successive transformations (Xue et al., 2024).

AGILE chemistry, an ester-bearing aldehyde supplies the ester functionality, so preparing that precursor is a separate synthesis problem from carrying out the final assembly (Xu et al., 2024). Repeated aza-Michael addition and reductive amination require the program to record the successive transformations and participating amine sites. A family name alone does not specify this complete assembly sequence.

Different precursor graphs can occupy the same reactive role because the assembly constraints specify reactive groups and their product connectivity. Functional regions such as headgroup, linker and tail need not coincide with precursor-origin boundaries: one precursor may contribute to several regions. An assembly description must therefore distinguish precursor origin, reactive role and transformation order from the molecular identity assigned to each role.

2.3 MOLECULAR GRAPHS AND REACTION PROGRAMS

A lipid constitution is a connected, labeled molecular graph $G = (V, E, \ell_V, \ell_E)$, where V and E are the atom and bond sets and ℓ_V and ℓ_E assign labels from finite atom and bond vocabularies. Bond order is an edge label. Molecular identity is constitutional and stereo-free: graphs are compared up to label-preserving isomorphism after removing atom-map and stereochemical labels, while retaining formal charges and bond orders. The declared support bounds the heavy-atom count and cycle rank. Atom maps remain available separately as evidence of precursor origin; they do not change the identity used for novelty or exact reconstruction.

A reaction program collects these assembly constraints in

$$P = (c, d, \mathbf{r}, \mathbf{k}, M), \quad (1)$$

where c identifies the ordered chemistry, d is its depth, $\mathbf{r}$ assigns precursor roles, and $\mathbf{k}$ marks reaction-core positions. For each role, M specifies the exterior atom count, junction budget, cycle rank and core-attachment count. These quantities constrain coarse architecture without specifying a precursor identity or stored subgraph. Precursor origin and core membership are separate: an atom can belong to a precursor and participate in the reaction core, whereas a template-introduced atom belongs to the core without being assigned to a precursor. Thus a program can constrain the assembly while leaving the surrounding molecular structures to be generated.

2.4 EXACT ASSEMBLY, NOVELTY AND ROUTE EVIDENCE

Exact assembly requires recovered precursors to reconstruct the generated graph. Let $\mathcal{F}_P(\mathbf{C})$ contain the registered forward products of precursor input $\mathbf{C}$, and let $\mathcal{D}_P(G)$ contain candidate pairs $(\mathbf{C}, \tau)$ from reverse search, where τ is the atom-mapped reaction trace. The verified set is

$$\hat{\mathcal{E}}_P(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}), H \cong G\}. \quad (2)$$

A valid graph passes verified exact level-one assembly (L1) when $\hat{\mathcal{E}}_P(G) \neq \emptyset$. Every admitted trace replays exactly. Incomplete reverse search can miss a valid decomposition. Coverage and replay precision therefore require separate definitions (Appendix A.5).

Verifier-recovered component novelty means that a verified trace contains a precursor absent from its role-specific training catalogue. This does not prove that the product is unreachable from known components: another decomposition could use only catalogue precursors. The experiments do not establish strong product-level catalogue escape (Appendix C.3).

A complete computational synthesis dossier requires L1 assembly together with evidence-qualified precursor preparation (L2) and terminal-material availability (L3). Incomplete assessments remain unresolved or search-censored. Neither exact assembly nor dossier closure establishes experimental synthesis success or delivery activity.

3 FORGE

3.1 OVERVIEW

FORGE generates a complete ionizable-lipid graph G conditioned on a reaction program P . The program specifies assembly chemistry, precursor roles, reaction-core positions, transformation order and coarse morphology. The conditional distribution $p_\theta(G | P)$ is learned without precursor identifiers, stored component graphs or fragment tokens. Generation and synthesis assessment proceed through four stages (Figure 1):

1. **Reaction-program specification.** P defines the assembly chemistry, precursor roles, reaction-core positions, transformation order, and coarse per-role morphology, leaving precursor identities unspecified.
2. **Whole-graph generation.** A conditional discrete flow generates the complete atom, bond, and topology state of G under P , preserving the program-fixed coordinates.
3. **Exact assembly verification.** A reverse transform decomposes G into role-assigned precursors, and the corresponding forward transform must reconstruct G exactly.
4. **Recursive synthesis-program construction.** FORGE searches for preparation routes to generated precursors that are not already supported as available starting materials and abstains when a route cannot be completed.

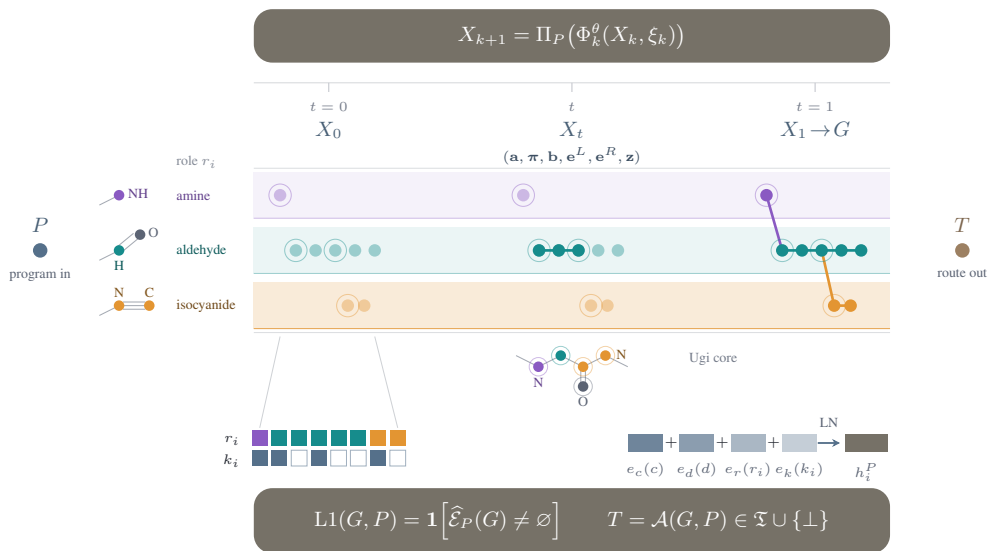


Figure 1: **FORGE generation and route assessment.** A reaction program fixes assembly roles and core coordinates without selecting precursor identities. FORGE generates the whole graph, verifies exact assembly replay, and recursively traces its implied precursors to evidence-qualified materials or an explicit abstention. Colours mark precursor origin; grey denotes the Ugi template-introduced oxygen. Replay and route closure are computational checks, not experimental synthesis evidence.

3.2 CONDITIONING ON REACTION PROGRAMS

We sample an admissible complete program P from a training-fold program prior and generate G under the resulting program,

$$p_{\theta}(G, P | c) = p_{\text{prog}}(P | c) p_{\theta}(G | P). \quad (3)$$

At graph position i , the program embedding is

$$h_i^P = \text{LN} \left(e_c(c) + e_d(d) + e_r(r_i) + e_k(k_i) + \sum_{u=1}^4 e_{M,u}(M_{r_i,u}) \right). \quad (4)$$

Here, e_c, e_d, e_r, e_k and $e_{M,u}$ are learned embeddings in a shared hidden space, and LN denotes layer normalization. The embedding encodes precursor role, reaction-core status and role-local morphology. The morphology fields determine active positions and fixed or variable masks. Sampling from the prior specifies these constraints; generation assigns the atom, bond and topology states within each role. Appendix A.1 specifies the morphology counts and adapter coordinates.

3.3 GRAPH REPRESENTATION AND DISCRETE FLOW

FORGE represents a complete molecule using a categorical tree-plus-closure state. Given the selected exact atom-mapped annotation α , the deterministic program-aware encoder assigns

$$X_G := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}), \quad (5)$$

where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ contains parent pointers, $\mathbf{b}$ contains spanning-tree bond states and $(\mathbf{e}^L, \mathbf{e}^R, \mathbf{z})$ encodes closure bonds. The encoder fixes the root, vertex order, spanning tree and closure order as specified in Appendix A.2. The decoder is surjective onto $\mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}}$, with $N_{\text{max}} = 194$ atoms and at most $K_{\text{cl}} = 3$ closure bonds. Every graph within the declared atom, bond and size support therefore has a valid representation, although representability does not imply reaction consistency or nonzero model probability; malformed raw states decode to $\perp_{\text{dec}}$.

Let $\mathcal{V}(P)$ and $\mathcal{F}(P)$ denote the variable and fixed coordinate sets. The program-compatible source q_0^P fixes $\mathcal{F}(P)$ and factorizes across $\mathcal{V}(P)$, with marginal $q_{0,i}^P$ at variable coordinate i . Let Ω_P be the raw states satisfying the fixed coordinates. Given a clean endpoint $x_1 \in \Omega_P$, the conditional path is

$$q_{t|1}^P(x | x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{1,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1-t)q_{0,i}^P(x_i) + t\mathbf{1}\{x_i = x_{1,i}\}]. \quad (6)$$

Each variable-coordinate distribution interpolates linearly between its source marginal and clean endpoint; the joint conditional law is the product of these distributions. A local continuous-time generator changes at most one coordinate infinitesimally, so the predicted clean-coordinate marginals determine its joint rate. Sampling begins from $X_0 \sim q_0^P$ and applies a capped independent-coordinate Euler approximation. Appendix C defines the rates and numerical kernel; the implementation does not claim exact continuous-time simulation.

3.4 PROGRAM-CONDITIONED TRANSFORMER AND TRAINING

FORGE applies this discrete flow to X_G and conditions endpoint prediction on P . A chemistry adapter converts the reaction program into masks over atom, pointer and bond coordinates. Program-fixed coordinates remain constant while the flow generates all remaining states. A graph Transformer predicts the clean atom, bond, pointer and closure states from (X_t, t, P) . Each layer uses self-attention with relation biases for parent, child and closure edges, then cross-attends to tokens encoding program identity, depth, precursor role and reaction-core position. Repeating this cross-attention in every layer makes program information available during both local updates and coordination between precursor-derived regions. Softly program-routed residual adapters add family-specific parameters to the shared whole-graph backbone. We supervised role and core annotations using qualified atom-mapped traces and excluded records with multiple unresolved source traces from semantic training.

Let $\mathcal{J}$ denote the six state families, $m_j^{(q)}(P) \in \{0, 1\}$ indicate whether coordinate j in family q remains variable, and $\ell_j^{(q)}(\theta) = -\log p_{\theta}(X_{G,j}^{(q)} | X_t, t, P)$ denote the corresponding denoising loss.

The primary masked objective is

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G,P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0,1), X_t \sim q_{t|1}^P(\cdot|X_G)} \left[\sum_{q \in \mathcal{J}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \ell_j^{(q)}(\theta)}{\max\{1, \sum_j m_j^{(q)}(P)\}} \right], \quad (7)$$

where μ_{bal} assigns equal total training mass to each reaction program and preserves source-balanced weights within each program, λ_q weights family q , and a family with no variable coordinates contributes zero loss. We added auxiliary role and core losses to each reaction family’s objective and combined the family task gradients using the projecting conflicting gradients (PCGrad) update (Yu et al., 2020). We used the deterministic task order specified in Appendix A.3. The population result applies only to the displayed primary term.

3.5 SAMPLING AND EXACT ASSEMBLY VERIFICATION

During corruption and generation, fixed coordinates have zero update rate and the projection Π_P restores them after every finite step. Numerical sampling therefore follows

$$X_{k+1} = \Pi_P(\Phi_k^\theta(X_k, \xi_k)), \quad k = 0, \dots, L_{\text{samp}} - 1, \quad (8)$$

where Φ_k^θ is the capped Euler update and ξ_k contains its sampling randomness. Program-fixed coordinates therefore remain constant throughout sampling. Chemical consistency with the requested assembly is established separately by exact forward replay. The terminal state is decoded into G and must be connected, valence-valid and chemically sanitized. Appendix C gives the representation and ideal population result, which excludes auxiliary losses, PCGrad, finite capacity and numerical integration error. Appendix A.4 specifies the finite-step sampler and terminal core-saturation constraints.

After generation, we recover and verify the implied precursors using Equation 2. This test verifies the specified final reaction without requiring the generator to select from a stored precursor catalogue. Molecular validation, candidate decomposition, exact replay and root ambiguity are recorded separately. The yield denominator includes every generation attempt, including failures at any stage.

3.6 RECURSIVE ROUTE ASSESSMENT

For precursors that are not directly available, a bounded program constructor searches for preparation routes and checks terminal supplier evidence,

$$Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}, \quad (9)$$

where $\mathfrak{D}$ is the dossier space, $\mathfrak{F}_{\mathfrak{B}}$ is the finite set of typed outcomes under the frozen route contract $\mathfrak{B}_{\text{route}}$, and $(\mathbf{C}_T, \tau_T)$ is the sole verifier-returned exact root trace. Ambiguous roots cause an abstention. A dossier is route-complete only when G passes molecular and exact-assembly verification, each upstream reaction is admitted and verified, and every required precursor traces to an available starting material. Otherwise the constructor returns an explicit unresolved or search-censored outcome.

Generation and route assessment remain separate. The reaction program P conditions the distribution of G , whereas $\text{Assess}_{\mathfrak{B}}$ evaluates the completed graph without modifying it. Appendix A specifies graph serialization, model training, numerical sampling and evidence admission, with the complete inference procedure in Algorithm 1. Appendix C states the assumptions and proofs of the formal results.

4 RESULTS

4.1 EVALUATION PROTOCOL AND MATCHED BUDGETS

We evaluated FORGE on Ugi, repeated aza-Michael addition and repeated reductive amination using three matched arms: shared conditioning, shared training without program information and shared training with cyclically incorrect reaction labels. All arms used the same architecture, optimization budget, attempt count and prespecified final checkpoint. We evaluated three independent training seeds. Verified exact-L1 yield is the fraction of generation attempts that passed molecular validation,

program-aware decomposition and exact forward replay. We excluded component-disjoint held-out products from training and checkpoint selection. Training seeds were the independent units for across-run comparisons. Appendix B contains the full protocol and seed-level results.

Equal attempt budgets account for generation failures. For a fixed model and program, let p be the probability that one attempt passes molecular validation and verified exact-L1 assembly. Under independent sampling, the accepted count A_B among B attempts satisfies

$$A_B \sim \text{Binomial}(B, p), \quad \mathbb{E}[A_B] = Bp. \quad (10)$$

Post-hoc filtering selects accepted outputs but does not change the proposal’s p . We therefore included every generation attempt in the denominator, including invalid and unverifiable outputs, and reported coverage separately from replay precision. The binomial law describes sampling variation for a fixed model. We summarized retraining variation using the three independent seed-level results. Proposition C.11 gives the corresponding accepted-count and sampling-cost results.

4.2 SHARED GENERATION ACHIEVED EXACT ASSEMBLY ACROSS THREE PROGRAMS

We first tested whether one conditional flow could learn all three reactions while preserving Ugi performance. Across 3,072 held-out attempts per program and seed, FORGE achieved verified exact-L1 yields of $96.4 \pm 2.1\%$ for Ugi, $72.2 \pm 1.6\%$ for repeated aza-Michael addition and $52.3 \pm 3.4\%$ for repeated reductive amination (Table 2). Every verifier-returned trace passed exact forward replay in every program and seed, as required by construction. Reductive amination showed greater between-seed variation and higher decomposition ambiguity, indicating less consistent learning in this lower-data setting.

FORGE produced no fixed-coordinate violations or support overflows across 27,648 held-out attempts. These runs established exact assembly across all three programs, with the highest yield for Ugi.

Table 2: **Exact assembly under three ionizable-lipid reaction programs.** Exact-L1 yields are mean $\pm$ sample standard deviation across three seeds. The final column reports decomposition coverage and the forward-replay verifier invariant. Per-seed values: Appendix B.

Program	FORGE conditioned	Shared null	Cyclic program	Decomp. coverage / replay invariant (%)
Ugi	96.4 ± 2.1	34.8 ± 7.1	93.8 ± 4.2	99.9/100.0
Aza-Michael	72.2 ± 1.6	19.2 ± 4.1	71.5 ± 2.2	100.0/100.0
Reductive amination	52.3 ± 3.4	37.0 ± 2.2	50.7 ± 4.9	86.6/100.0

4.3 PROGRAM CONDITIONING IMPROVED EXACT-ASSEMBLY YIELD

To isolate the effect of reaction information during generation, we compared the conditioned model with the shared-null and cyclic arms under the same attempt budget and post-hoc verifier. Shared generation without program information yielded $34.8 \pm 7.1\%$, $19.2 \pm 4.1\%$ and $37.0 \pm 2.2\%$ exact L1 for Ugi, aza-Michael addition and reductive amination, respectively (Table 2). For Ugi, conditioning increased yield by 61.6 percentage points (three-seed range 57.4 to 67.7 points).

Assigning the wrong reaction identity reduced exact-L1 yield to $93.8 \pm 4.2\%$, $71.5 \pm 2.2\%$ and $50.7 \pm 4.9\%$ across the three programs. The cyclic control retained precursor roles, reaction-core coordinates and program depth, isolating the contribution of the reaction-identity label within a structured program. The larger conditioned-versus-null differences supported a benefit from program information during generation. The small conditioned-versus-cyclic differences did not establish a strong independent contribution from the identity label.

4.4 DECODER AND ARCHITECTURE EFFECTS DIFFERED ACROSS PROGRAMS

Exact assembly depends on both the learned distribution and terminal decoding. In a seed-0 diagnostic with trained weights held fixed, enforcing the qualified Ugi core’s hydrogen states increased exact-L1 yield relative to valence-only topology decoding, while molecular validity was nearly unchanged

(Appendix B.3). The core-saturation constraint changed the generated graph, and we applied the same exact-replay verifier to both decoders. Molecular validity alone therefore did not account for the improvement in reaction consistency. Changing the initial corruption distribution from program-role marginals to global marginals also reduced yield. These fixed-model diagnostics complement the independently retrained comparisons in Table 2.

A separate three-seed mechanism study compared whole-graph denoising with a parameter-matched component-factorized model (FACT), which generates precursor roles independently (Appendix B.3). Its full-Transformer control was trained independently of the production model. Mean exact-L1 products per 1,000 attempts were 856.9 versus 56.9 for Ugi and 767.7 versus 396.4 for aza-Michael addition. The direction reversed for reductive amination: 318.8 versus 520.3. Increasing FACT capacity did not reverse these mean orderings. Joint denoising improved exact assembly in two programs. The negative conditional-dependence diagnostic left the mechanism of this advantage unresolved.

Individual modules likewise had mixed effects. Removing layerwise program cross-attention reduced Ugi and reductive-amination yield but increased aza-Michael yield; removing gradient-conflict control slightly improved all three means. The ablations did not support a uniform benefit from every module across the tested reaction programs.

4.5 GENERATED PRECURSORS EXTENDED BEYOND THE TRAINING CATALOGUES

We next asked whether FORGE could preserve exact assembly while generating precursor constitutions absent from training. Under the common Ugi benchmark, the finite-inventory methods produced 1000.0 ± 0.0 verified exact-L1 products per 1,000 attempts but zero component-novel products by construction. FORGE produced 963.9 ± 21.4 verified products and 862.6 ± 19.2 distinct component-novel products per 1,000 attempts (Table 3).

Table 3: Exact Ugi assembly with precursor constitutions absent from method-visible training. Values are mean $\pm$ sample standard deviation per 1,000 attempts across three seeds under a common component-disjoint protocol and verifier. Component-novel is verifier-relative and does not establish strong product-level catalogue escape. Whole-molecule baselines are assessed post hoc. $\dagger$, structural zero; N/E, not estimable.

Models	Valid / 1,000 $\uparrow$	Verified L1 / 1,000 $\uparrow$	Distinct L1 / 1,000 $\uparrow$	Comp.-novel / 1,000 $\uparrow$	Held-comp. / 1,000 $\uparrow$	Diversity
RGFN	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	$0.0 \pm 0.0^\dagger$	0.0 ± 0.0	N/E
DeFoG unconditional	346.5 ± 42.3	280.6 ± 26.6	280.6 ± 26.6	280.6 ± 26.6	15.5 ± 2.4	0.727 ± 0.014
GenMol/SAFE	627.5 ± 6.2	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.873 ± 0.009
Learned inventory selector	1000.0 ± 0.0	1000.0 ± 0.0	849.5 ± 13.7	$0.0 \pm 0.0^\dagger$	0.0 ± 0.0	0.717 ± 0.001
Finite catalogue oracle	1000.0 ± 0.0	1000.0 ± 0.0	837.6 ± 8.2	$0.0 \pm 0.0^\dagger$	0.0 ± 0.0	0.719 ± 0.000
Shared-null + post-hoc	194.7 ± 15.6	114.4 ± 10.9	108.5 ± 9.7	108.5 ± 9.7	0.4 ± 0.2	0.754 ± 0.014
FACT-matched	60.8 ± 20.2	52.0 ± 15.6	50.8 ± 14.4	50.8 ± 14.4	0.0 ± 0.0	0.667 ± 0.009
FACT-generous	500.7 ± 213.8	296.7 ± 186.0	288.3 ± 175.3	288.3 ± 175.3	0.1 ± 0.2	0.774 ± 0.007
FORGE Transformer	965.0 ± 21.9	963.9 ± 21.4	862.6 ± 19.2	862.6 ± 19.2	1.4 ± 1.7	0.643 ± 0.022

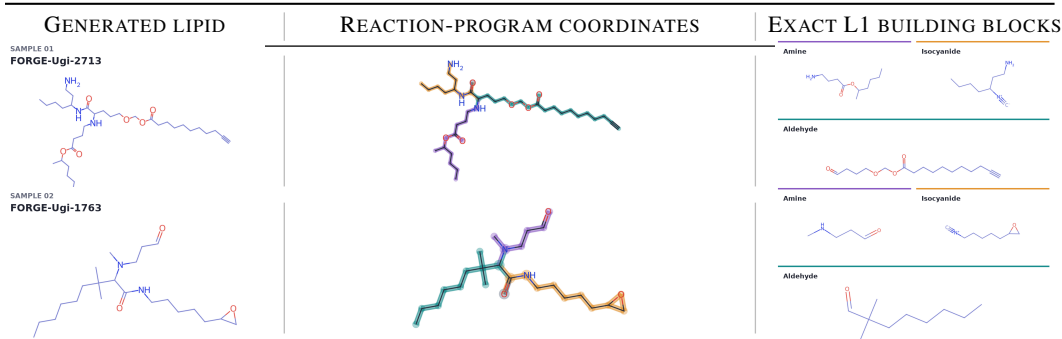
Against the fixed training catalogues, component-novel verified exact-L1 yields reached 839.2, 159.3 and 58.7 products per 1,000 attempts across Ugi, aza-Michael addition and reductive amination. The corresponding finite catalogues had zero component-novel yield by construction (Appendix B.4). Representative Ugi products are shown in Figure 2.

Among the external baselines, DeFoG produced 280.6 ± 26.6 verified products and recovered held components more often than FORGE (15.5 ± 2.4 versus 1.4 ± 1.7 per 1,000 attempts). GenMol/SAFE produced no exact Ugi-L1 product, and RGFN emitted no admissible output. FORGE therefore combined reaction consistency with verifier-relative component novelty, although DeFoG recovered the reserved components more often. Product-level catalogue escape remained unmeasured.

Component novelty also did not establish recovery of the held-out lipid distribution. The method-blind realism analysis measured both precision and coverage in fingerprint and descriptor spaces, together with a source-study-grouped classifier two-sample test (Appendix B.4). No evaluated method reproduced the full held-out observed-lipid distribution. The measured expansion was limited to

reaction-consistent molecular identities. Neither diversity nor distance from training established agreement with observed lipid structures or biological utility.

Figure 2: Exact Ugi products with recovered precursors absent from training. Each row contains a deterministic seed-0 product with a unique verified exact-L1 decomposition and a verifier-recovered precursor absent from training. Columns show the generated graph, atom-mapped origins and recovered precursors; halos mark the Ugi core and colours encode atom origin. These are structural examples, not evidence of activity or experimental synthesis success.



4.6 PRECURSOR EVIDENCE LIMITED COMPLETE ROUTE CONSTRUCTION

To assess whether the verified assemblies had complete computational synthesis dossiers, we checked precursor preparation and terminal-material evidence in addition to exact L1 replay. Only 30 of 11,021 components in the frozen evidence union closed under the exact-source library. FORGE therefore produced 963.9 ± 21.4 verified exact-L1 products per 1,000 attempts but 0.0 ± 0.0 complete dossiers. In a route-blind sample of 256 products, 0 closed, 256 were unresolved and 0 were search-censored (Appendix B.5). These outcomes measured completion under the frozen evidence policy. The syntheses of unresolved products remain unknown.

5 DISCUSSION

FORGE conditions whole-molecule generation on known assembly chemistry with precursor identities left open. Across the three specified programs, conditioning increased verified exact-L1 yield relative to matched generation followed by post-hoc verification. Generated products also contained verifier-recovered precursors absent from the training catalogues. The combination of complete molecular generation and exact assembly replay permits exploration beyond the supplied precursor identities without requiring a finite component inventory in the generative state.

The evidence supports component-level expansion under the registered chemistry. Complete catalogue-product reachability was not computed, so the results do not establish strong product-level catalogue escape. Held-component recovery was weak (1.4 ± 1.7 products per 1,000 attempts), and no evaluated method reproduced the full held-out lipid distribution. Joint denoising and individual architectural modules also had reaction-dependent effects. Higher exact-assembly yield therefore did not imply uniform gains in generalization, structural realism or every reaction program.

Route assessment further bounded the demonstrated capability. Exact final assembly does not establish precursor availability, and the bounded evidence library left the final-model products without complete dossiers. Neither synthesis guidance nor the bounded potency diagnostic improved the matched outcome (Appendix B). The study is computational, covers three programs within bounded graph support, and includes no prospective synthesis, formulation or biological validation. Evidence was strongest for Ugi; extending qualified precursor evidence and testing additional supported programs would address the limits of route completion and chemical coverage.

AI USE STATEMENT

Generative AI assistance was used to adapt the manuscript’s LaTeX formatting, file organization and bibliography presentation to the ICLR template, and to reorganize and expand the main text using existing appendix material and cited literature. AI assistance was also used for the full-manuscript language revision, citation checks and consistency review.

REPRODUCIBILITY STATEMENT

The model is defined in Section 3; encoding, training, sampling and verification details are specified in Appendix A. Experimental protocols, evaluation definitions, extended results and seed-level tables are provided in Appendix B. Assumptions and proofs of the formal claims appear in Appendix C.

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A METHOD AND IMPLEMENTATION DETAILS

This appendix specifies program and graph encoding, model training, sampling, verification and route assessment. Algorithm 1 combines the inference steps.

A.1 REACTION-PROGRAM ENCODING

Definition A.1 (Reaction program). *A reaction program is a tuple*

$$P = (c, d, \mathbf{r}, \mathbf{k}, M), \quad (11)$$

with an ordered layout index set $\mathcal{I}(P) = \{1, \dots, n(P)\}$. Here c identifies the ordered assembly chemistry, d is the program depth, $\mathbf{r} = (r_i)_{i \in \mathcal{I}(P)}$ assigns precursor roles, $\mathbf{k} = (k_i)_{i \in \mathcal{I}(P)}$ marks reaction-core positions, and M specifies coarse per-role morphology. For each precursor role r , M records four integers: the number of exterior atoms n_r , junction budget j_r , cycle rank γ_r , and attachment count a_r .

M is defined relative to an exact atom-mapped trace τ and precursor tuple $\mathbf{C}$. An atom is exterior for role r when τ assigns it to the corresponding precursor and does not mark it as reaction core. Template-introduced atoms belong to the core and to no precursor role. The role-induced subgraph contains the atoms assigned to r and all product bonds with both endpoints in that set. Its junction count is $\sum_v \max\{\deg_r(v) - 2, 0\}$, its cycle rank is $|E_r| - |V_r| + \kappa_r$ for κ_r connected components, and its core-attachment count is the number of product bonds between an exterior atom of r and a core atom. We write $(G, \mathbf{C}, \tau) \models M(P)$ when these values match the sampled counts, with junction count at most its budget. Morphology therefore depends on the selected decomposition of G .

FORGE first samples an admissible complete program P from the training-fold program prior and then generates G conditioned on it:

$$p_\theta(G, P \mid c) = p_{\text{prog}}(P \mid c) p_\theta(G \mid P). \quad (12)$$

The prior sets each region’s atom count, branching budget, cycle rank, and number of core attachments, but not its molecular structure. A chemistry-specific adapter defines the precursor roles, reaction core, reverse decomposition, and forward transform associated with c . For each graph position i , FORGE combines learned embeddings of the program identity, program depth, precursor role, and reaction-core position:

$$h_i^P = \text{LN} \left(e_c(c) + e_d(d) + e_r(r_i) + e_k(k_i) + \sum_{u=1}^4 e_{M,u}(M_{r_i,u}) \right). \quad (13)$$

The morphology fields determine active positions and fixed/variable masks in the sampling layout. The resulting coordinates encode each position’s precursor role, reaction-core status and role-local counts without specifying its molecular structure.

A.2 GRAPH SERIALIZATION AND SOURCE ANNOTATIONS

A spanning-tree representation with at most three residual closure edges preserves the full 194-atom support without an $O(N_{\text{max}}^2)$ edge state. The discrete flow generates the atom, bond and topology states in this representation. Let $\alpha = (\mathbf{C}, \tau, \mathbf{r}, \mathbf{k})$ contain the selected exact precursor tuple, atom-mapped reaction trace, precursor-role assignment and reaction-core annotation. The deterministic program-aware encoder

$$\text{Enc}_P : \{(G, \alpha) : G \in \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \text{ and } \alpha \text{ is admitted for } P\} \longrightarrow \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \quad (14)$$

constructs the categorical graph state

$$X_G := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}) \quad (15)$$

where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ parent pointers, $\mathbf{b}$ spanning-tree bond states, $\mathbf{e}^L$ and $\mathbf{e}^R$ closure-endpoint pointers, and $\mathbf{z}$ closure-bond states.

The encoder first partitions the atoms into connected precursor-origin components under α . It selects as root the reaction-core atom with the smallest tie-broken canonical RDKit rank, and orders the origin-component graph by breadth-first traversal from that root component, breaking ties by role index, component canonical ranks and original atom index. Within each component it uses a deterministic depth-first preorder from the selected attachment atom, with neighbors ordered by the same canonical ranks. The traversed bonds define the spanning tree. Every residual bond is represented once as an ordered endpoint pair with its bond label, and residual triples are sorted lexicographically. Role, core, fixed-coordinate and morphology arrays follow the same atom order. We excluded training records with more than one qualified exact source trace from semantic training and collapsed duplicate descriptions of an admitted trace to its canonical record. Thus $\text{Dec}(\text{Enc}_P(G, \alpha)) \cong G$ for every admitted record.

The raw padded categorical product space $\bar{\mathfrak{X}}_{N_{\max}, K_{\text{cl}}}$ also contains malformed states. Its well-formed subset $\mathfrak{X}_{N_{\max}, K_{\text{cl}}}$ enforces an active node prefix, earlier-parent pointers, distinct valid closure endpoints, no duplicate edges, and null values in unused closure slots. The partial decoder is therefore typed as

$$\text{Dec} : \bar{\mathfrak{X}}_{N_{\max}, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N_{\max}, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (16)$$

For this study, $N_{\max} = 194$ and $K_{\text{cl}} = 3$. These bounds specify architectural support. Reaction consistency, nonzero probability under a trained model and synthesis-program completeness require separate conditions.

If $q_{\theta}(x \mid P)$ is the terminal raw-state law, the generated graph distribution is its decoder pushforward,

$$p_{\theta}([G]_{\cong} \mid P) = \sum_{x: \text{Dec}(x) \cong G} q_{\theta}(x \mid P), \quad p_{\theta}(\perp_{\text{dec}} \mid P) = \sum_{x: \text{Dec}(x) = \perp_{\text{dec}}} q_{\theta}(x \mid P). \quad (17)$$

The chemistry adapter marks any atom, pointer, or bond states fixed by P . FORGE holds these states constant while corrupting the remaining states along the linear probability path and predicting their clean values from (X_t, t, P) . Let Ω_P be the face of the raw state space satisfying those fixed coordinates, and define the admissible program set

$$\mathcal{P}_{\text{adm}} = \{P : \mathfrak{X}_{N_{\max}, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}. \quad (18)$$

The morphology prior is normalized only on $\mathcal{P}_{\text{adm}}$. Each chemistry adapter is required to satisfy the program-compatible encoding property: every admitted annotated record obeys $\text{Enc}_P(G, \alpha) \in \Omega_P$. During corpus construction, we checked this property and exact constitutional decode round trip for every record before split assignment. Records failing either check were excluded. The checks covered all cached training, calibration and held-out records; they do not establish the property for unregistered chemistry.

A.3 TRANSFORMER ARCHITECTURE AND TRAINING OBJECTIVES

The denoiser is a graph Transformer over the complete sparse state. Each layer applies graph self-attention with relation biases for parent, child and closure edges, then cross-attends to memory tokens representing program identity, program depth, precursor role and reaction-core position. Cross-attention is repeated in every layer so that reaction semantics can influence both local state updates and long-range coordination among precursor-derived regions. Three softly program-routed residual adapters add family-specific parameters to the shared whole-graph backbone.

Let $\mathcal{J} = \{\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}\}$ denote the six state families optimized during training. Let μ_{bal} denote the frozen training distribution over graph-program pairs, which assigns equal total mass to each reaction program and preserves the source-balanced weights within each program. For each $q \in \mathcal{J}$, let $m_j^{(q)}(P) \in \{0, 1\}$ indicate whether coordinate j remains variable under P . We trained the denoiser using the program-balanced objective over variable coordinates:

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G, P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0, 1), X_t \sim q_{t|1}^P(\cdot | X_G)} \left[\sum_{q \in \mathcal{J}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \left[-\log p_{\theta}(X_{G,j}^{(q)} \mid X_t, t, P) \right]}{\max\{1, \sum_j m_j^{(q)}(P)\}} \right]. \quad (19)$$

Here, λ_q weights state family q , and a family with no variable coordinates contributes zero loss. The corruption and sampling operators assign zero rates to program-fixed coordinates and project each finite-step state back onto them. These operations keep fixed coordinates constant throughout the trajectory. Chemical consistency with the requested assembly is established separately by exact forward replay. Under an unrestricted coordinatewise model class, every population minimizer of the primary masked denoising term recovers the corresponding clean-coordinate posterior marginals. This result does not characterize the optimizer of the auxiliary-loss or PCGrad procedure (Propositions C.6 and C.7).

For each reaction family f , we compute a task loss

$$\mathcal{L}_f = \mathcal{L}_{\text{FORGE},f} + \lambda_{\text{role}}\mathcal{L}_{\text{role}} + \lambda_{\text{core}}\mathcal{L}_{\text{core}}. \quad (20)$$

We assigned equal mass to present semantic states within each auxiliary term and equal total loss mass to every reaction family. Deterministic PCGrad combines the gradients of the family task losses after summing their primary and auxiliary terms. Families are ordered by their integer program state. Writing $g_f = \nabla_{\theta}\mathcal{L}_f$, the algorithm initializes $\tilde{g}_f \leftarrow g_f$ and, for every other family h in that fixed order, applies

$$\tilde{g}_f \leftarrow \tilde{g}_f - \mathbf{1}\{\langle \tilde{g}_f, g_h \rangle < 0\} \frac{\langle \tilde{g}_f, g_h \rangle}{\max(\|g_h\|_2^2, 10^{-12})} g_h \quad (21)$$

on parameter coordinates available to both tasks. Zero gradients remain zero; gradients are not norm-normalized; and the optimizer receives the arithmetic mean of the projected family gradients. This is an algorithmic definition of the update and is not the gradient of the displayed scalar sum in general. We tested layerwise cross-attention, each auxiliary loss, routed adapters and gradient-conflict control through separate interventions (Appendix B.3).

A.4 NUMERICAL SAMPLING AND TERMINAL DECODING

At inference, FORGE samples X_0 from the factorized program-compatible source, assigns zero update rates to fixed coordinates, and applies L_{samp} capped independent-coordinate Euler steps. At each step the implementation samples one clean category per coordinate from the denoiser, forms the minimum R^* rate for that sampled endpoint, multiplies it by $\Delta t = 1/L_{\text{samp}}$, and rescales the total off-diagonal mass to at most 0.999 before drawing the next coordinate value. The resulting sampler approximates a local continuous-time Markov chain (CTMC) through finite steps and Monte Carlo endpoint draws. Appendix C states the ideal marginalized rate and the numerical kernel. FORGE decodes the terminal state X_1 into a molecular graph G and retains it only if it is connected, valence-valid, and passes molecular sanitization.

For Ugi, the production decoder also enforces the qualified core’s saturation states: one hydrogen on the aldehyde-derived α carbon and one on the isocyanide-derived amide nitrogen. These constraints affect the generated graph; they do not change the subsequent exact-replay criterion. We compared core-saturation and valence-only topology decoding under the same trained weights and verifier, then separately compared program-role and global source marginals (Appendix B.3).

A.5 EXACT ASSEMBLY VERIFICATION

To ensure that each retained graph satisfies the specified reaction program, FORGE decomposes it into role-assigned precursors and applies the corresponding forward transform.

Definition A.2 (Mathematical and verifier-relative L1). *For chemistry c , the reverse search $\mathcal{D}_c$ maps G to candidate role-assigned precursor tuples and atom-mapped traces. The complete mathematical preimage $\mathcal{E}_P^*(G)$ contains every admissible input whose registered forward operator can produce G (Appendix C.3). Define*

$$\text{L1}^*(G, P) = \mathbf{1}[\mathcal{E}_P^*(G) \neq \emptyset]. \quad (22)$$

The bounded implementation searches $\mathcal{D}_P(G) \subseteq \mathcal{D}_c(G)$ and returns

$$\hat{\mathcal{E}}_P(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}) \text{ such that } H \cong G\}. \quad (23)$$

Its verified predicate is

$$\widehat{\text{L1}}(G, P) = \mathbf{1}[\hat{\mathcal{E}}_P(G) \neq \emptyset]. \quad (24)$$

Every returned trace passes exact replay, hence $\widehat{L1}(G, P) \leq L1^*(G, P)$. Reverse-search completeness has not been proved, so the reported metric is called *verified exact-L1 yield*. Post-hoc filtering changes which completed samples are retained but cannot change the proposal distribution’s exact-L1 acceptance probability. Under independent sampling, this probability determines the complete distribution of accepted counts under a fixed attempt budget and the expected cost of obtaining a fixed number of accepted products (Proposition C.11). We distinguish decomposition coverage from forward-replay precision because a candidate disconnection may fail to reconstruct G . Candidate decomposition coverage is the fraction of valid graphs for which $\mathcal{D}_c(G) \neq \emptyset$. Candidate replay precision is

$$\text{ReplayPrecision} = \frac{\sum_G \sum_{(C, \tau) \in \mathcal{D}_P(G)} \mathbf{1}\{\exists H \in \mathcal{F}_P(C) : H \cong G\}}{\sum_G |\mathcal{D}_P(G)|}, \quad (25)$$

with N/E when the denominator is zero. Replay of tuples already in $\widehat{\mathcal{E}}_P(G)$ has 100% success by construction. This invariant does not estimate candidate replay precision. We separately define

$$\text{CandidateAmbig}(G, c) = \mathbf{1}\{|\mathcal{D}_c(G)| > 1\}, \quad \text{ExactAmbig}(G, P) = \mathbf{1}\{|\widehat{\mathcal{E}}_P(G)| > 1\}. \quad (26)$$

A distinct exact-L1 product is a deduplicated generated constitution. An unambiguous exact decomposition satisfies $|\widehat{\mathcal{E}}_P(G)| = 1$. We distinguish product uniqueness from decomposition uniqueness throughout.

A.6 RECURSIVE ROUTE ASSESSMENT AND DOSSIER ADMISSION

A.6.1 ASSESSMENT INTERFACE

For a generated molecular graph G and reaction program P , route assessment returns either a complete synthesis dossier or a typed abstention. Let $\mathcal{D}$ denote the dossier space and let $\mathfrak{F}_{\mathfrak{B}}$ be the finite set of typed failure outcomes under a frozen route contract $\mathfrak{B}_{\text{route}}$. Generation and assessment are defined by

$$G \sim p_{\theta}(\cdot | P), \quad Y = \text{Assess}_{\mathfrak{B}}(G, P, C_T, \tau_T) \in \mathcal{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (27)$$

$\text{Assess}_{\mathfrak{B}}$ is invoked after the verifier returns one admitted root trace $(C_T, \tau_T) \in \widehat{\mathcal{E}}_P(G)$. It verifies the final assembly, searches for preparation routes to unavailable precursors and checks terminal supplier evidence. P and the assessor act at different stages: P guides generation, whereas the assessor constructs a dossier from the completed graph without modifying it. FORGE accepts (G, T) only when G passes molecular and exact-assembly verification and T traces every required precursor to an available starting material; otherwise, it abstains.

A.6.2 RECURSIVE EXPANSION AND COMPLETION CHECKS

Completing a synthesis dossier requires evidence for every recovered precursor. Exact L1 verification specifies the precursor structures; availability and preparation routes require additional evidence. Each precursor must therefore match an available material or have a supported route from available starting materials. Let $T = (V_M, V_R, E)$ denote a bipartite synthesis tree rooted at G , where each product molecule points to its preparation reaction and each reaction points to its required reactants. The assessor constructs these routes recursively and withholds a complete dossier when any branch remains unresolved. A molecule node terminates when a current, dated supplier record lists an exact constitutional match as available; otherwise, FORGE must expand that node through a supported, forward-verified reaction. FORGE uses a bounded level-two (L2) planner to propose each expansion. For each proposal, FORGE records the reaction, reactants, product, and supporting evidence, and accepts the step only when the forward transform reconstructs the target precursor exactly. Each accepted reaction adds its reactants as new molecule nodes, and FORGE applies the same termination-or-expansion rule recursively. The current assessor admits only an unambiguous verified exact-L1 root. Every dossier therefore records the sole selected root trace $(C_T, \tau_T) \in \widehat{\mathcal{E}}_P(G)$; an ambiguous exact set produces a typed abstention. For each upstream reaction node r , let $\text{Admitted}_{\mathfrak{B}}(r) = 1$ when its reaction and substrate scope satisfy the frozen evidence policy, and let $\text{Verify}_{\mathfrak{B}}(r) = 1$ when $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$. For each leaf molecule ℓ , let $\text{Available}_{\mathfrak{B}}(\ell) = 1$ when it satisfies the terminal supplier-evidence rule. Let $V_R^{\text{up}}(T)$ denote the upstream preparation-reaction nodes and

$L(T)$ the leaf molecules. $\text{WellFormed}(G, T, P)$ requires that T is a connected, acyclic bipartite tree rooted at G ; each internal molecule has exactly one recorded producing reaction; each reaction’s molecule children equal its recorded reactants; and T has no disconnected, cyclic or extraneous node. Its distinguished final-assembly record must have product G and precursor children exactly C_T under trace τ_T .

$$\begin{aligned} \text{Complete}_{\mathfrak{B}}(G, T, P) = & \text{WellFormed}(G, T, P) \mathbf{1} \left[(C_T, \tau_T) \in \widehat{\mathcal{E}}_P(G) \right] \\ & \times \prod_{r \in V_R^{\text{up}}(T)} \text{Admitted}_{\mathfrak{B}}(r) \text{Verify}_{\mathfrak{B}}(r) \prod_{\ell \in L(T)} \text{Available}_{\mathfrak{B}}(\ell). \end{aligned} \quad (28)$$

Empty products equal one. Let $\mathfrak{B}_{\text{route}}$ fix the route-depth and query budgets, registered transforms, evidence-admission policy, and dated supplier snapshot used by $\text{Assess}_{\mathfrak{B}}$. Under finite search bounds and terminating evidence queries, the assessor terminates and returns a dossier only when $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$ (Lemma C.12 and Invariant C.13). An abstention means that FORGE did not complete a dossier under $\mathfrak{B}_{\text{route}}$; it does not imply that no synthesis route exists outside the frozen search space or evidence snapshot. The guarantee is computational and does not establish experimental synthesis success.

A.6.3 RECORDED EVIDENCE AND SUPPORT TIERS

For every accepted pair (G, T) , FORGE records a computational synthesis dossier containing all molecules and reactions in T , the evidence supporting each reaction, and the dated supplier record for every terminal material. We classify each generated lipid into one of four support tiers according to which elements of its assembly lie outside the fixed reference space: **E0 (enumerated)**: the generated lipid matches a product that was already constructed computationally from known components using a supported reaction. **E1 (known components)**: the generated lipid was not included in the bounded E0 enumeration, but a supported reaction assembles it entirely from known components. **E2 (generated precursor)**: a supported reaction assembles the lipid, but at least one precursor is newly generated and requires a complete L2 route to available starting materials. **E3 (new assembly family)**: the lipid requires a final assembly reaction outside the qualified reaction registry.

A.7 END-TO-END INFERENCE PROCEDURE

The inference procedure records every attempt and its outcome (Algorithm 1). Each iteration consumes one generation attempt, including those that fail molecular validation, exact replay or route completion.

Algorithm 1 FORGE generation and fail-closed synthesis-program construction.

Require: trained coordinate predictors, reaction program P , Euler-step count L_{samp} , attempt budget B , recorded seed s , and frozen route contract $\mathfrak{B}_{\text{route}}$

Ensure: attempt-indexed dossier sequence $\mathcal{D}_{\text{out}}$, exact-L1 ledger $\mathcal{L}_{\text{out}}$, and typed failure ledger $\mathcal{H}$

```

1: initialize the random-number streams from  $s$ ;  $\mathcal{D}_{\text{out}} \leftarrow []$ ;  $\mathcal{L}_{\text{out}} \leftarrow []$ ;  $\mathcal{H} \leftarrow []$ 
2: for  $b = 1, \dots, B$  do
3:   sample  $X_0 \sim q_0^P$  and set  $X \leftarrow \Pi_P(X_0)$ 
4:   for  $k = 0, \dots, L_{\text{samp}} - 1$  do
5:      $t_k \leftarrow k/L_{\text{samp}}$ ;  $\Delta t_k \leftarrow 1/L_{\text{samp}}$ 
6:      $X \leftarrow \Pi_P(\text{CAPPEDCOORDINATEEULER}(X, t_k, \Delta t_k, P))$ 
7:   end for
8:    $G \leftarrow \text{DECODE}(X)$ 
9:   if  $G = \perp_{\text{dec}}$ , is valence-invalid, or fails molecular sanitization then
10:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{INVALIDGRAPH})]$ 
11:    continue
12:   end if
13:    $\hat{\mathcal{E}}_P(G) \leftarrow \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}), H \cong G\}$ 
14:   if  $\hat{\mathcal{E}}_P(G) = \emptyset$  then
15:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{L1FAILURE})]$ 
16:    continue
17:   end if
18:    $\mathcal{L}_{\text{out}} \leftarrow \mathcal{L}_{\text{out}} \parallel [(b, G, \hat{\mathcal{E}}_P(G))]$ 
19:   if  $|\hat{\mathcal{E}}_P(G)| \neq 1$  then
20:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{AMBIGUOUSL1})]$ 
21:    continue
22:   end if
23:   select the sole  $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_P(G)$ 
24:    $Y \leftarrow \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$ 
25:   if  $Y$  is a tree  $T$  and  $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$  then
26:     $\mathcal{D}_{\text{out}} \leftarrow \mathcal{D}_{\text{out}} \parallel [(b, \text{DOSSIER}(G, P, \mathbf{C}_T, \tau_T, T))]$ 
27:   else
28:     $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, Y)]$  ▷ preserve the typed outcome
29:   end if
30: end for
31: return  $\mathcal{D}_{\text{out}}, \mathcal{L}_{\text{out}}, \mathcal{H}$ 

```

B EVALUATION PROTOCOL AND ADDITIONAL RESULTS

The evaluation protocol applies to Section 4. Additional analyses quantify decomposition outcomes, variation across training seeds, component diversity and the bounded guidance diagnostics. Each comparison includes its seed-level results and metric definitions.

B.1 EVALUATION PROTOCOL

B.1.1 DATA, SPLITS AND GENERATION BUDGETS

We evaluated three assembly programs with different amounts of supervision: Ugi, repeated aza-Michael addition and repeated reductive amination. The comparisons tested shared generation, reaction conditioning, recovery of precursors absent from training and completion of synthesis dossiers. We trained three matched shared-model arms: correct program conditioning, no program information and cyclically incorrect reaction labels. All arms used the same architecture and optimization budget. We evaluated the prespecified final checkpoint from three independent training seeds under equal generation-attempt budgets per program. Our primary metric is verified exact-L1 yield per generation attempt: the number of generated graphs that pass molecular validity, program-aware decomposition, and exact forward replay divided by the total number of sampling attempts. Secondary metrics separate graph quality (validity and connectedness), reaction recovery (verified coverage, the replay invariant, abstention, and exact ambiguity), and exploration (distinct-product fraction, whole-product and component novelty, molecular diversity, and effective component count). We report exact counts, means across seeds, sample standard deviations, individual seed values, and the minimum and maximum of the three paired seed contrasts. These are descriptive across-seed summaries, not confidence intervals. Generated molecules are not treated as independent training replicates. Conditional binomial variation across a fixed model’s 3,072 attempts is distinct from variation induced by retraining. Training used only the component-disjoint training folds. We used calibration data for diagnostics and excluded held-out products from training and checkpoint selection. All three shared arms assigned equal total sampling mass to each program and preserved the frozen within-program source weights. We also checked the prespecified numerical acceptance criteria. With three training seeds, these checks describe the observed runs and do not establish population-level non-inferiority. Zero program-fixed-coordinate violations are an implementation check required by Proposition C.6, not learned behavior.

B.1.2 COMPARATORS AND MATCHED CONTROLS

The null and cyclic controls tested different aspects of conditioning. Null generation followed by exact post-hoc filtering measured performance without program information. The cyclic arm isolated the reaction-identity label with the remaining program structure retained. A separate train-only finite-component assembler received the exact training components, supported forward chemistry, source-weighted role marginals and the same number of attempts. We evaluated reaction validity and exploration separately because catalogue assembly has zero input-component novelty by construction. None of these comparisons called an L2 planner, biological oracle or candidate-selection procedure.

For Ugi, we evaluated a common external-baseline protocol with one reaction-space comparator, RGFN, and two generic whole-molecule comparators, unconditional DeFoG and GenMol/SAFE (Kozierski et al., 2024; Qin et al., 2025; Lee et al., 2025). The Synthesis-DAG method of Ou et al. (2024), SynFlowNet and SynCoGen were discussed as related methods and were excluded from the empirical comparison (Cretu et al., 2025; Rekesh et al., 2026). At the version checks used to construct the benchmark, the Synthesis-DAG repository lacked the paper-specific ionizable-lipid implementation, the SynFlowNet action space did not support the qualified three-reactant Ugi transform, and the SynCoGen repository lacked a usable code license. We preserved each method’s native reaction operations and included numerical results only from executions using the common training split, attempt budget, seed set and molecular and exact-L1 evaluator. The comparison excludes published scores from other datasets and unsupported aza-Michael or reductive-amination runs. The matched internal suite included unconditioned generation plus post-hoc filtering, an oracle finite-catalogue assembler, a learned inventory selector, two learned component-factorized generators, and targeted interventions on the Transformer. Appendix B.3 reports the three-seed Transformer

mechanism and component-factorization study. Each intervention used the same folds, generation budget and evaluation code as the full Transformer.

B.1.3 METRICS, MOLECULAR IDENTITY AND UNCERTAINTY

Let $Y_{s,a,b}^{L1} \in \{0, 1\}$ indicate verified exact L1 for training seed s , arm a and attempt b , and let $B = 3,072$. We report the exact count $C_{s,a}^{L1} = \sum_{b=1}^B Y_{s,a,b}^{L1}$ and

$$\hat{p}_{s,a}^{L1} = C_{s,a}^{L1}/B. \quad (29)$$

Validity and connected yield replace Y^{L1} with their attempt-level indicators. Verified decomposition coverage is the number of valid graphs with $\hat{\mathcal{E}}_P(G) \neq \emptyset$ divided by valid graphs; the implementation does not report candidate-search coverage before replay. Exact ambiguity is the fraction of valid graphs with $|\hat{\mathcal{E}}_P(G)| > 1$. Replay of members of $\hat{\mathcal{E}}_P(G)$ is an invariant, while candidate replay precision is defined in Appendix A and is N/E if no candidate tuple is returned. Explicitly, with attempt outcomes $G_{s,a,b}$ and the convention that a failed decode makes every graph predicate zero,

$$\hat{p}_{s,a}^{\text{valid}} = B^{-1} \sum_b \mathbf{1}\{\text{Valid}(G_{s,a,b}) = 1\}, \quad (30)$$

$$\hat{p}_{s,a}^{\text{conn}} = B^{-1} \sum_b \mathbf{1}\{\text{Valid}(G_{s,a,b}) = 1, \text{Connected}(G_{s,a,b}) = 1\}, \quad (31)$$

$$\text{CandidateCoverage} = \frac{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1, \mathcal{D}_c(G_b) \neq \emptyset\}}{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1\}}, \quad (32)$$

$$\text{ExactAmbiguity} = \frac{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1, |\hat{\mathcal{E}}_P(G_b)| > 1\}}{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1\}}. \quad (33)$$

Each ratio is N/E when its denominator is zero.

Distinct-product yield is the number of distinct canonical product constitutions satisfying verified exact L1 divided by B . The reported verifier-recovered component-novel yield additionally requires $|\hat{\mathcal{E}}_P(G)| = 1$ and at least one returned role-assigned precursor constitution absent from the fixed training catalogue. It is not StrongEscape. Held-component recovery counts verified exact-L1 attempts whose returned tuple contains a designated held constitution and divides by B . Writing $\text{can}(G)$ for constitutional canonicalization, $\mathcal{C}_{\text{train}}$ for the role-indexed training catalogue and $\mathcal{H}_{\text{held}}$ for the designated held constitutions, these yields are

$$\text{DistinctL1Yield} = B^{-1} \left| \left\{ \text{can}(G_b) : \widehat{\text{L1}}(G_b, P) = 1 \right\} \right|, \quad (34)$$

$$\begin{aligned} \text{ComponentNovelYield} = B^{-1} \sum_b \mathbf{1} \Big[& \widehat{\text{L1}}(G_b, P) = 1, |\hat{\mathcal{E}}_P(G_b)| = 1, \\ & \exists(r, C) \in \mathbf{C}_b : C \notin \mathcal{C}_{\text{train},r} \Big], \end{aligned} \quad (35)$$

$$\text{HeldComponentYield} = B^{-1} \sum_b \mathbf{1} \Big[\widehat{\text{L1}}(G_b, P) = 1, \mathbf{C}_b \cap \mathcal{H}_{\text{held}} \neq \emptyset \Big]. \quad (36)$$

For component frequencies w_i , the effective count is

$$N_{\text{eff}} = \exp \left(- \sum_i w_i \log w_i \right). \quad (37)$$

Component duplicates are retained when estimating w_i . Molecular diversity is the mean pairwise ECFP4 Tanimoto distance among distinct valid canonical product constitutions, so molecular duplicates are removed for that metric. For distinct products $g_1, \dots, g_m$,

$$\text{Diversity} = \frac{2}{m(m-1)} \sum_{1 \leq i < j \leq m} [1 - \text{Tanimoto}(\text{ECFP4}(g_i), \text{ECFP4}(g_j))], \quad (38)$$

with N/E for $m < 2$. Every other undefined conditional denominator is likewise reported N/E.

Identity is constitutional and stereo-free. Graph and product identity use connected canonical RDKit SMILES with atom maps and stereochemical tags removed; formal charges and bond orders are retained; salts or disconnected products are invalid; and no tautomer normalization is applied. Precursor identity uses the same canonicalization. Precursor-tuple equality is role- and program-step-aware, with repeated inputs compared in their registered step order. Atom mappings and the trace τ are evidence for semantic alignment, not part of constitutional identity.

For a paired contrast, $D_s = \hat{p}_{s,\text{FORGE}} - \hat{p}_{s,\text{control}}$. We report (D_1, D_2, D_3) , $\bar{D}$, and $[\min_s D_s, \max_s D_s]$. These summarize the three observed training seeds and are not 95% confidence intervals. Conditional on one fixed trained model, $C_{s,a}^{\text{L1}}$ has the finite-attempt binomial variation in Proposition C.11; that uncertainty does not substitute for independent retraining. Unless explicitly named primary, architecture, coordinate, guidance, potency and catalogue contrasts are exploratory and no multiplicity-adjusted inferential claim is made.

B.2 ASSEMBLY QUALITY AND PROGRAM CONDITIONING

B.2.1 SHARED GENERATION ACROSS THREE CHEMISTRIES

The full shared-model comparison used 3,072 held-out generation attempts per program in each of three independent training seeds (Table 4). Table 2 summarizes the exact-L1 yields. Every verifier-returned trace passed forward replay, as required by the verifier invariant. Mean exact- decomposition ambiguity was 61.0% for repeated aza-Michael addition and 74.7% for repeated reductive amination, and the latter program showed substantial between-seed variation. Table 4 reports molecular validity, connectedness, decomposition coverage, forward-replay precision, abstention, ambiguity, novelty and diversity separately for each program.

The final conditioned model recorded zero fixed-coordinate violations and zero support overflows across 27,648 held-out attempts (three seeds, three programs and 3,072 attempts per program). Exact-assembly yields differed across reaction families despite preservation of all fixed coordinates and graph-support bounds.

Table 4: **Matched shared-model production comparison across reaction programs.** Point estimates report the mean across three independent seeds; the exact-L1 column additionally reports the sample standard deviation. Paired contrasts are summarized by their three seed values, mean and observed range. Conditioned-model counts by seed appear in Table 5. Validity and connectedness use all generation attempts as the denominator. Decomposition coverage, replay precision, abstention and ambiguity retain their prespecified conditional denominators. Novelty is conditioned on valid products or unique returned decompositions, as appropriate. Fraction pairs and exact-L1 yield are percentages; diversity is mean pairwise ECFP4 distance and N_{eff} is the effective generated-component count. Results are never pooled across reaction programs.

	Ugi			Aza-Michael			Reductive amination		
	FORGE	Shared null	Cyclic	FORGE	Shared null	Cyclic	FORGE	Shared null	Cyclic
Valid./Conn.	96.5/96.5	35.9/35.9	93.9/93.9	72.2/72.2	27.4/27.4	71.5/71.5	60.4/60.4	41.9/41.9	59.1/59.1
Decomp./Replay	99.9/100.0	97.0/100.0	99.8/100.0	100.0/100.0	69.5/100.0	100.0/100.0	86.6/100.0	88.4/100.0	85.7/100.0
Abst./Ambig.	0.1/0.0	3.0/0.9	0.2/0.0	0.0/61.0	30.5/29.3	0.0/62.3	13.4/74.7	11.6/76.6	14.3/73.8
Exact-L1/attempt	96.4 $\pm$ 2.1	34.8 $\pm$ 7.1	93.8 $\pm$ 4.2	72.2 $\pm$ 1.6	19.2 $\pm$ 4.1	71.5 $\pm$ 2.2	52.3 $\pm$ 3.4	37.0 $\pm$ 2.2	50.7 $\pm$ 4.9
Whole/Comp. novelty	99.5/96.7	99.9/99.0	99.4/96.9	89.8/85.3	99.6/99.7	90.2/86.3	67.4/97.5	91.5/99.8	66.0/97.0
Diversity/ N_{eff}	0.643/419.9	0.680/439.5	0.646/427.9	0.593/118.0	0.707/413.5	0.603/119.4	0.776/173.0	0.761/193.3	0.773/157.8

B.2.2 CONDITIONED, NULL AND CYCLIC CONTROLS

Under the matched protocol in Table 4, the paired conditioned-minus-null differences were 61.6 percentage points for Ugi (three-seed range 57.4 to 67.7), 53.0 for aza-Michael (50.2 to 56.4), and 15.4 for reductive amination (14.5 to 16.7).

The cyclic control retained precursor roles, reaction-core coordinates and program depth but assigned the wrong reaction identity. Relative to correct conditioning, this intervention changed verified exact-L1 yield by 2.6 percentage points for Ugi (three-seed range 0.7 to 5.5), 0.6 for aza-Michael (-2.4 to 3.8), and 1.6 for reductive amination (-0.1 to 2.5). Every returned exact trace passed replay by verifier construction.

Table 5: **Exact verified-L1 counts for the final conditioned model.** Each row reports the numerator, fixed attempt denominator and resulting percentage for one independently trained seed. These attempt-level counts quantify conditional Monte Carlo variation; the three training seeds are the independent units for retraining variation. Seed labels 0, 1 and 2 correspond to RNG seeds 20260825, 20260826 and 20260827, respectively.

Program	Seed	Verified L1 count	Attempts	Yield (%)
Ugi	0	2,922	3,072	95.12
Ugi	1	3,037	3,072	98.86
Ugi	2	2,924	3,072	95.18
Aza-Michael	0	2,176	3,072	70.83
Aza-Michael	1	2,203	3,072	71.71
Aza-Michael	2	2,274	3,072	74.02
Reductive amination	0	1,561	3,072	50.81
Reductive amination	1	1,727	3,072	56.22
Reductive amination	2	1,535	3,072	49.97

We independently retrained the null and cyclic arms under the same study protocol to measure how program information affected learned generation. Correct conditioning improved yield substantially relative to the null arm. Its smaller advantage over cyclic labels did not establish a strong independent contribution from the reaction-identity token.

B.3 DECODER AND ARCHITECTURE ABLATIONS

B.3.1 TERMINAL DECODING AND SOURCE MARGINALS

We evaluated terminal decoding and source marginals in separate fixed-weight comparisons (Table 6).

Table 6: **Decoder and source-marginal ablations of the final model.** Ugi held-out attempts for the 9,143-step shared production model. The two ablations are independent: the decoder rows share one set of trained weights and differ only in terminal decoding, while the source rows share one decoder and differ only in the marginal the corruption starts from. Decoding under the reaction-core saturation policy holds the aldehyde-derived α -carbon at one hydrogen and the isocyanide-derived amide nitrogen at one hydrogen, which the qualified transform fixes; it changes the generated graph under an unchanged verifier. Values are percentages from 3,072 seed-0 held-out attempts per row; they are descriptive and do not measure across-training-seed variation. Exact-L1 coverage is conditioned on valid products.

Source marginal	Terminal decoder	Valid	Exact-L1 coverage	Exact-L1
Program-role	Core saturation	95.21%	99.90%	95.12%
Program-role	Valence topology only	95.31%	78.35%	74.67%
Global	Core saturation	86.95%	99.63%	86.62%

B.3.2 JOINT DENOISING, ROLE ISOLATION AND TRANSFORMER MODULES

The matched-study full Transformer produced 856.9 Ugi and 767.7 repeated aza-Michael exact-L1 products per 1,000 attempts, compared with 56.9 and 396.4 for the parameter-matched role-isolated FACT model (Table 7). The full Transformer and FACT-matched model each had 5.33 million trainable parameters; FACT-generous had 29.40 million. The higher-capacity FACT control improved to 455.0 for Ugi and 685.7 for repeated aza-Michael addition. Its three-seed means remained below the full Transformer for both programs, although the aza-Michael values were essentially tied in one seed. The direction reversed for repeated reductive amination: the full Transformer produced 318.8 exact-L1 products per 1,000 attempts, whereas FACT-matched and FACT-generous produced 520.3 and 558.1. Thus, joint whole-lipid denoising improved exact assembly for Ugi and repeated aza-Michael addition but was not uniformly superior across reaction programs. Because the prespecified conditional-dependence diagnostic was negative, this architecture contrast does not establish detectable cross-role dependence in the present data.

Table 7: **Transformer architecture and mechanism ablations.** The five Transformer intervention rows change only the named mechanism from the matched-study full model; the two FACT rows generate precursor roles independently at matched or increased capacity. All arms use identical data folds, corruption, context schedule, checkpoint rule, core-saturation terminal decoder and sampling budget. The reference full Transformer is the matched mechanism-study control, not the independently trained production model reported in Table 4 and Appendix B.4. Entries report the mean $\pm$ sample standard deviation across three independent training seeds. Held-out loss is the mean denoising loss at $t = 0.5$ across the three component-disjoint program partitions. Cross-role fidelity is the Frobenius distance to the held-out residualized descriptor matrix, for which lower values indicate closer agreement. Exact-L1 yields use all 3,072 attempts per program and seed as the denominator.

Arm	Held-out loss	Ugi L1/1k	BL L1/1k	LX L1/1k	Held-comp. L1/1k	Cross-role fidelity	Diversity
Input-only program	8.48 $\pm$ 0.72	795.2 $\pm$ 112.4	776.5 $\pm$ 33.2	268.1 $\pm$ 56.9	0.00 $\pm$ 0.00	1.83 $\pm$ 0.15	0.597 $\pm$ 0.009
No role loss	7.87 $\pm$ 0.12	846.2 $\pm$ 98.2	769.6 $\pm$ 45.6	289.7 $\pm$ 148.1	0.22 $\pm$ 0.19	1.92 $\pm$ 0.07	0.587 $\pm$ 0.011
No core loss	7.85 $\pm$ 0.11	833.4 $\pm$ 101.7	776.0 $\pm$ 37.2	310.1 $\pm$ 130.1	0.22 $\pm$ 0.19	1.82 $\pm$ 0.15	0.584 $\pm$ 0.007
No routed adapters	7.81 $\pm$ 0.32	791.0 $\pm$ 164.9	791.0 $\pm$ 49.5	329.9 $\pm$ 60.9	0.11 $\pm$ 0.19	1.77 $\pm$ 0.07	0.587 $\pm$ 0.001
No gradient conflict control	7.98 $\pm$ 0.20	866.4 $\pm$ 84.1	774.1 $\pm$ 44.2	324.1 $\pm$ 157.4	0.11 $\pm$ 0.19	1.92 $\pm$ 0.12	0.584 $\pm$ 0.008
FACT-matched	7.09 $\pm$ 0.24	56.9 $\pm$ 24.4	396.4 $\pm$ 63.5	520.3 $\pm$ 62.2	0.11 $\pm$ 0.19	1.31 $\pm$ 0.22	0.555 $\pm$ 0.014
FACT-generous	8.24 $\pm$ 0.31	455.0 $\pm$ 191.9	685.7 $\pm$ 107.4	558.1 $\pm$ 63.6	1.95 $\pm$ 0.86	1.12 $\pm$ 0.04	0.589 $\pm$ 0.005
FORGE	7.88 $\pm$ 0.22	856.9 $\pm$ 86.7	767.7 $\pm$ 35.9	318.8 $\pm$ 158.3	0.11 $\pm$ 0.19	1.90 $\pm$ 0.13	0.585 $\pm$ 0.007

Removing layerwise program cross-attention reduced mean exact-L1 yield by 61.6 products per 1,000 attempts for Ugi and 50.7 for repeated reductive amination, while increasing repeated aza-Michael yield by 8.8. Removing the role or core objective and removing routed adapters also produced program-dependent changes. The arm without gradient-conflict control slightly exceeded the full Transformer mean in all three programs. These interventions did not support a uniform benefit from every module. We reported effects separately by chemistry and did not combine them into an architecture score.

B.4 COMPONENT NOVELTY, GENERALIZATION AND STRUCTURAL REALISM

B.4.1 COMMON UGI BENCHMARK

We compared FORGE with external generators, matched internal controls and the finite-catalogue oracle under the common Ugi benchmark (Table 3). Every yield is normalized by the number of generation attempts, including attempts that fail before producing a valid molecule. All included methods were run under the common split, attempt budget, seed set and verifier.

The learned finite-inventory selector achieved 1000.0 ± 0.0 verified exact-L1 products per 1,000 attempts, but its finite action space imposed zero verifier-recovered component-novel yield. FORGE produced 963.9 ± 21.4 verified exact-L1 products and 862.6 ± 19.2 distinct verified exact-L1 products per 1,000 attempts. Among whole-molecule external generators, unconditional DeFoG produced 280.6 ± 26.6 verified exact-L1 products per 1,000 attempts and recovered held components at 15.5 ± 2.4 per 1,000 attempts, compared with 1.4 ± 1.7 for FORGE. GenMol/SAFE produced 627.5 ± 6.2 valid connected outputs but no exact Ugi-L1 product, while RGFN emitted no admissible molecular output under the strict native contract. These results established different performance orderings for reaction consistency, verifier-recovered component novelty and recovery of reserved components. They do not establish product-level strong catalogue escape.

Seed-level results and decomposition outcomes are reported in Tables 8 and 9. All values derive from hash-pinned three-seed results. We assessed L2/L3 evidence separately (Appendix B.5). Compute records are retained in the generated evidence receipt, but missing training-time fields prevent a cross-method compute-matching claim.

B.4.2 CATALOGUE COMPARISON AND COMPONENT DIVERSITY

At the matched sampling budget, FORGE produced 839.2 verifier-recovered component-novel Ugi products per 1,000 attempts, averaged across the three independent seeds. The corresponding values were 159.3 for aza-Michael addition and 58.7 for reductive amination. The finite-component assembler produced zero by construction because every sampled product uses only components from

Table 8: **Seed-level common Ugi benchmark results.** Each method has a hash-verified three-seed result vector. Table 3 reports the corresponding mean $\pm$ sample standard deviation. Training seeds are the independent units; generated molecules are samples from each fixed model.

Method	Seed(s)	Valid/1k	Verified L1/1k	Distinct L1/1k	Comp.-novel/1k	Held-comp. L1/1k	Diversity
RGFN	20260825	0.0	0.0	0.0	0.0	0.0	N/E
RGFN	20260826	0.0	0.0	0.0	0.0	0.0	N/E
RGFN	20260827	0.0	0.0	0.0	0.0	0.0	N/E
DeFoG unconditional	20260825	299.2	251.0	251.0	251.0	13.7	0.711
DeFoG unconditional	20260826	359.7	288.4	288.4	288.4	18.2	0.733
DeFoG unconditional	20260827	380.5	302.4	302.4	302.4	14.6	0.736
GenMol/SAFE	20260825	627.3	0.0	0.0	0.0	0.0	0.883
GenMol/SAFE	20260826	633.8	0.0	0.0	0.0	0.0	0.869
GenMol/SAFE	20260827	621.4	0.0	0.0	0.0	0.0	0.866
Learned inventory selector	20260825	1000.0	1000.0	863.0	0.0	0.0	0.717
Learned inventory selector	20260826	1000.0	1000.0	835.6	0.0	0.0	0.717
Learned inventory selector	20260827	1000.0	1000.0	849.9	0.0	0.0	0.715
Finite catalogue oracle	20260825	1000.0	1000.0	841.5	0.0	0.0	0.719
Finite catalogue oracle	20260826	1000.0	1000.0	843.1	0.0	0.0	0.719
Finite catalogue oracle	20260827	1000.0	1000.0	828.1	0.0	0.0	0.719
Shared-null + post-hoc	20260825	183.9	114.3	109.7	109.7	0.3	0.738
Shared-null + post-hoc	20260826	187.5	103.5	98.3	98.3	0.7	0.763
Shared-null + post-hoc	20260827	212.6	125.3	117.5	117.5	0.3	0.761
FACT-matched	20260825	59.2	49.5	48.8	48.8	0.0	0.676
FACT-matched	20260826	81.7	68.7	66.1	66.1	0.0	0.658
FACT-matched	20260827	41.3	37.8	37.4	37.4	0.0	0.668
FACT-generous	20260825	391.0	161.8	161.1	161.1	0.3	0.782
FACT-generous	20260826	747.1	508.8	488.3	488.3	0.0	0.768
FACT-generous	20260827	363.9	219.4	215.5	215.5	0.0	0.771
FORGE Transformer	20260825	952.1	951.2	872.4	872.4	3.3	0.656
FORGE Transformer	20260826	990.2	988.6	875.0	875.0	1.0	0.656
FORGE Transformer	20260827	952.5	951.8	840.5	840.5	0.0	0.618

Table 9: **Reaction decomposition outcomes under the common Ugi verifier.** Verified decomposition coverage is conditioned on molecular validity. Replay of every tuple in the returned exact set is a verifier invariant, not an estimate of reverse-search precision. Abstention and exact ambiguity retain their prespecified denominators. N/E denotes a conditional metric with no eligible valid product or returned decomposition.

Method	Valid products	Verified coverage	Replay invariant	Abstention	Exact ambiguity
RGFN	0.0 $\pm$ 0.0	N/E	N/E	N/E	N/E
DeFoG unconditional	1064.3 $\pm$ 129.9	81.2 $\pm$ 2.4	100.0 $\pm$ 0.0	18.8 $\pm$ 2.4	0.0 $\pm$ 0.0
GenMol/SAFE	1927.7 $\pm$ 19.0	0.0 $\pm$ 0.0	N/E	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Learned inventory selector	3072.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Finite catalogue oracle	3072.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Shared-null + post-hoc	598.0 $\pm$ 47.9	58.8 $\pm$ 3.5	100.0 $\pm$ 0.0	41.2 $\pm$ 3.5	0.0 $\pm$ 0.0
FACT-matched	186.7 $\pm$ 62.1	86.3 $\pm$ 4.4	100.0 $\pm$ 0.0	13.7 $\pm$ 4.4	0.0 $\pm$ 0.0
FACT-generous	1538.0 $\pm$ 656.9	56.6 $\pm$ 13.7	100.0 $\pm$ 0.0	43.4 $\pm$ 13.7	0.0 $\pm$ 0.0
FORGE Transformer	2964.3 $\pm$ 67.3	99.9 $\pm$ 0.0	100.0 $\pm$ 0.0	0.1 $\pm$ 0.0	0.0 $\pm$ 0.0

the training catalogue (Table 10). The catalogue achieved exact-L1 yields of 1000.0, 723.2 and 1000.0 products per 1,000 attempts, compared with 963.9, 721.9 and 523.3 for FORGE. The corresponding raw validities were 1000.0, 723.2 and 1000.0 for the catalogue and 965.0, 722.2 and 604.3 for FORGE. We report these quantities separately because reaction validity and verifier-relative component novelty measure different capabilities. Direct membership in the complete catalogue-reachable product set was not computed, so these results are not reported as strong catalogue escape.

In the distinct-product Ugi role analysis, 19,520 of 26,235 products (74.4%) contained at least one verifier-returned component constitution absent from the training registry. Across all Ugi products included in the novelty analysis, 21,960 of 26,235 distinct products (83.7%) were absent from the full 112,386-product constitutional reference corpus. FORGE effective component counts were 419.9, 118.0 and 173.0, compared with 86.4, 23.4 and 16.9 for the finite catalogue. The Ugi catalogue contained 185 amines, 75 aldehydes and 35 isocyanides and covered 100% of training products but 0% of the calibration and component-disjoint held-out products. Its Cartesian tuple ceilings were 485,625 for Ugi, 774 for aza-Michael and 140 for reductive amination; these count possible component-program inputs, whose forward products may overlap. The catalogue comparison therefore quantified both assembly consistency and the expansion of recovered precursor identities beyond that inventory.

Table 10: Whole-graph generation versus a finite-component assembler. Verified exact-L1, distinct exact-L1 and verifier-recovered component-novel yields are reported per 1,000 generation attempts. A component-novel product has exactly one verifier-returned exact-L1 decomposition and contains at least one component constitution absent from the training catalogue. This verifier-relative metric is not strong catalogue escape because complete catalogue-product reachability was not computed. The catalogue has zero input-component novelty by construction, but may achieve higher raw validity or exact-L1 yield. FORGE-minus-catalogue contrasts use the three seeds as descriptive paired units; no three-seed confidence interval is inferred.

	Ugi		Aza-Michael		Reductive amination	
	FORGE	Finite cat.	FORGE	Finite cat.	FORGE	Finite cat.
Valid/1k	965.0	1000.0	722.2	723.2	604.3	1000.0
Verified exact L1/1k	963.9	1000.0	721.9	723.2	523.3	1000.0
Distinct exact L1/1k	862.6	837.6	566.1	147.5	270.6	52.1
Verifier-recovered component-novel/1k	839.2	0.0	159.3	0.0	58.7	0.0
Whole novelty	99.5	16.9	89.8	39.8	67.4	44.8
Component novelty	96.7	0.0	85.3	0.0	97.5	0.0
Diversity/ N_{eff}	0.643/419.9	0.719/86.4	0.593/118.0	0.402/23.4	0.776/173.0	0.514/16.9

B.4.3 HELD-COMPONENT RECOVERY

The component-disjoint Ugi fold was never used for training, checkpoint selection or inventory construction. Under the common native-generation assessment, the full Transformer recovered only 13 exact-L1 products containing a designated held component in 9,216 attempts, or 1.4 ± 1.7 products per 1,000 attempts averaged across three seeds. Recovery of specifically reserved components was therefore weak despite the broader component novelty observed above. Novelty accepts any recovered precursor absent from training; held-component recovery requires a match to a designated withheld identity. The results do not establish strong held-component recovery, role-specific held-component coverage or zero-shot generalization to held reaction families.

B.4.4 STRUCTURAL REALISM

No evaluated method reproduced the full held-out observed-lipid distribution: where estimable, grouped C2ST AUC remained between 0.961 and 0.974, and fingerprint-manifold precision was zero or near zero (Table 11). GenMol/SAFE’s high pairwise diversity was paired with an effective count of only 3.2 ± 0.1 molecules, confirming severe frequency collapse. FORGE retained 2489.2 ± 66.8 effective molecules, but only 29.5 ± 2.6 attempts per 1,000 fell within the descriptor manifold. Diversity, effective molecule count and reference-distribution coverage therefore measured different aspects of the generated populations.

Table 11: **Method-blind structural-realism diagnostic.** Every entry is mean $\pm$ sample standard deviation across three training seeds and retains all 3,072 requested attempts in the denominator. FP and descriptor P/C report manifold precision per 1,000 attempts and coverage of the source-study-held-out R0 reference in percent. C2ST is grouped classifier two-sample area under the receiver operating characteristic curve; 0.5 denotes chance discrimination. N_{eff} is the Shannon effective molecule count. N/E denotes a metric that cannot be estimated. This post-hoc structural diagnostic is not a property, activity or synthesis-success score.

Method	Connected/1k	FP P/C	Desc. P/C	C2ST	N_{eff}	Diversity
RGFN	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	N/E	0.0 $\pm$ 0.0	N/E
DeFoG unconditional	346.5 $\pm$ 42.3	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	17.3 $\pm$ 1.4/1.45 $\pm$ 0.06	0.961 $\pm$ 0.007	1064.3 $\pm$ 129.9	0.726 $\pm$ 0.013
GenMol/SAFE	627.5 $\pm$ 6.2	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	N/E	3.2 $\pm$ 0.1	0.873 $\pm$ 0.009
Finite catalogue oracle	1000.0 $\pm$ 0.0	0.2 $\pm$ 0.2/0.05 $\pm$ 0.05	54.6 $\pm$ 4.4/2.02 $\pm$ 0.17	0.971 $\pm$ 0.007	2395.5 $\pm$ 31.7	0.720 $\pm$ 0.000
Learned inventory selector	1000.0 $\pm$ 0.0	0.1 $\pm$ 0.2/0.02 $\pm$ 0.03	56.6 $\pm$ 7.2/1.97 $\pm$ 0.46	0.974 $\pm$ 0.002	2443.0 $\pm$ 55.8	0.718 $\pm$ 0.001
FORGE Transformer	965.0 $\pm$ 21.9	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	29.5 $\pm$ 2.6/2.21 $\pm$ 0.03	0.972 $\pm$ 0.004	2489.2 $\pm$ 66.8	0.642 $\pm$ 0.022

B.5 ROUTE EVIDENCE AND DOSSIER CLOSURE

Table 12 reports each method’s finite-vocabulary constraint separately from evidence obtained under the common FORGE assessor. We froze the union of exact-L1 components across methods and seeds before route assessment. Method membership remained hidden until every component had a complete, unresolved or search-censored disposition. Emitting a reaction sequence is not counted as a complete dossier unless the final assembly and every upstream step pass exact forward verification and every terminal material closes under the frozen evidence policy. We excluded a separate AiZynthFinder row because a proposed route requires the same stepwise forward verification and terminal evidence before it qualifies as a complete dossier (Genheden et al., 2020).

The frozen union contained 11,021 distinct role–constitution components, but the exact-source library closed only 30. Consequently, FORGE produced 963.9 ± 21.4 verified exact-L1 products per 1,000 attempts, yet 0.0 ± 0.0 complete dossiers and 963.9 ± 21.4 abstentions under this library. The learned inventory selector closed 0.7 ± 0.3 dossiers per 1,000 attempts through occasional reuse of covered training components. The low completion counts reflected limited exact-source evidence coverage. Components outside the frozen library were not searched, leaving their synthesizability unknown.

Table 12: **Common synthesis-evidence assessment of Ugi generative methods.** The finite-vocabulary field records whether generation is restricted to a fixed component inventory. We applied the same method-blind component union, upstream forward-verification requirements, terminal-evidence rules and abstention policy to every method’s outputs. All numerical fields are reported per 1,000 generation attempts as mean $\pm$ sample standard deviation across three seeds. A zero downstream count means that no attempt reached that evidence state under the frozen policy. An abstention records missing exact evidence, not a failed synthesis; the frozen exact-source component library is deliberately non-exhaustive.

Method	Finite vocab.	Verified L1/1k	Verified upstream	Terminal evidence	Complete dossier	Abstention
RGFN	yes	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
DeFoG unconditional	no	280.6 $\pm$ 26.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	280.6 $\pm$ 26.6
GenMol/SAFE	no	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Finite catalogue oracle	yes	1000.0 $\pm$ 0.0	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	999.9 $\pm$ 0.2
Learned inventory selector	yes	1000.0 $\pm$ 0.0	0.7 $\pm$ 0.3	0.7 $\pm$ 0.3	0.7 $\pm$ 0.3	999.3 $\pm$ 0.3
Shared-null + post-hoc	no	114.4 $\pm$ 10.9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	114.4 $\pm$ 10.9
FACT-matched	no	52.0 $\pm$ 15.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	52.0 $\pm$ 15.6
FORGE Transformer	no	963.9 $\pm$ 21.4	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	963.9 $\pm$ 21.4

For the final-model assessment, we selected a deterministic, route-blind sample of 256 distinct products drawn by canonical-product hash from 7696 valid, verified exact-L1 products across the three final training seeds. All 256 products retained one exact forward-verified root trace. We assessed each product against the frozen evidence index. 0 products closed under the exact-source and terminal-evidence policy, 256 remained unresolved and 0 were censored. The products contained 379 unique role–constitution component targets: 4 complete, 375 unresolved and 0 censored. No planner or proposal engine was called during scoring (Table 13).

Table 13: **Route-evidence outcomes for final Transformer products.** The deterministic hash-ranked sample contains distinct Ugi products with one exact forward-verified L1 trace. Complete denotes closure of every recovered component under the frozen exact-source upstream-reaction and terminal-evidence policy. Unresolved and censored outcomes are abstentions, not evidence of unsynthesizability.

Unit	Population	Complete	Unresolved	Censored
Products	256	0	256	0
Unique components	379	4	375	0

B.6 GUIDANCE DIAGNOSTICS

At synthesis-guidance strength 0.25, the matched controller evaluated 64 terminal particles and retained 56 canonical representatives per arm. It changed only 8 terminal identities and returned 2 unique route-ready products, compared with 3 after post-hoc assessment. Only 5 of 48 four-particle morphology groups contained mixed route-readiness values, leaving most ancestry decisions uniform. This limited contrast did not establish a benefit from mid-trajectory synthesis guidance. The production method retained complete generation followed by proposal-augmented L1/L2/L3 route assessment.

We evaluated biological guidance only in the bounded HeLa messenger-transfection-potential (mTP) diagnostic. A group-disjoint preterminal classifier detected weak morphology-level signal, with ROC AUC 0.623 (95% interval 0.565 to 0.679) and a 1.177-fold observed top-quartile gain. We then compared a frozen applicability-enriched morphology proposal with a nested potency-enriched proposal using 3,072 generator calls per arm. Table 14 reports the matched terminal outcome.

Table 14: **Predicted HeLa messenger-transfection potential under matched generation budgets.** The support-enriched arm used the selected applicability proposal. The nested arm reallocated morphology mass using the frozen weak potency signal. Conservative-high counts are computational predictions, not transfection measurements. The comparison uses the same generator-call and oracle budgets.

Arm	Attempts	Verified L1	Eligible before budget	Oracle budget	Unique conservative-high
Support-enriched	3,072	2955	90	38	27
Nested potency proposal	3,072	2962	82	38	23

The nested-minus-support difference was -1.30 conservative-high products per 1,000 generator calls (95% interval -2.28 to 0.98). The comparison failed the prespecified advancement criterion. These computations evaluate predictor behavior within a declared applicability domain; they do not establish improved biological activity, delivery or experimental potency.

C THEORETICAL ANALYSIS AND PROOFS

We first establish the sparse representation’s correctness and bounded completeness, then characterize finite-component reachability. The discrete-flow results connect the primary denoising objective to the ideal conditional graph law under explicit population and exact-simulation assumptions. The final results quantify matched-budget filtering and bounded route assessment.

These results concern representation, population objectives, and computational verification. They do not imply that finite-data training assigns positive probability to every representable graph, that registered transformations cover arbitrary chemistry, that bounded route search is complete, or that a returned dossier guarantees experimental synthesis success.

C.1 NOTATION AND SETUP

Symbol	Meaning
$\Sigma_{\text{atom}}, \Sigma_{\text{bond}}$	atom-state and bond-state alphabets
N, K_{cl}	maximum active nodes and closure-edge budget
L_{samp}	finite Euler-step count
$\mathcal{I}$	six categorical state families
$\mathcal{F}(P), \mathcal{V}(P)$	fixed and variable coordinate sets
$\mathcal{C}_P$	admissible precursor-program inputs
$\mathfrak{B}_{\text{route}}$	frozen route-assessment contract
$\mathcal{D}, \mathcal{D}_{\text{out}}$	dossier space and attempt-indexed dossier sequence
Q	proposal law in the post-hoc comparison

A molecular graph is $G = (V, E, \ell_V, \ell_E)$, where the underlying graph is finite, connected, undirected and simple, $\ell_V : V \rightarrow \Sigma_{\text{atom}}$, and $\ell_E : E \rightarrow \Sigma_{\text{bond}}$. Bond multiplicity belongs to ℓ_E ; parallel edges are not permitted. Define the set of label-preserving isomorphism classes

$$\mathfrak{G}_{N, K_{\text{cl}}} = \{[G]_{\cong} : |V| \leq N, \beta(G) = |E| - |V| + 1 \leq K_{\text{cl}}, \ell_V(V) \subseteq \Sigma_{\text{atom}}, \ell_E(E) \subseteq \Sigma_{\text{bond}}\}. \quad (39)$$

Let $\bar{\mathfrak{X}}_{N, K_{\text{cl}}}$ be the full padded categorical product space. A state belongs to its well-formed subset $\mathfrak{X}_{N, K_{\text{cl}}}$ exactly when:

1. for some $1 \leq n \leq N$, the active nodes are exactly $\{1, \dots, n\}$, every active node has a label in Σ_{atom} , and inactive node slots carry the null state;
2. node 1 is the unique root and carries the null parent-edge sentinel;
3. every $i > 1$ has $\pi_i \in \{1, \dots, i-1\}$ and a parent-edge label in Σ_{bond} ;
4. every active closure slot joins two distinct endpoints in $\{1, \dots, n\}$, carries a label in Σ_{bond} , and duplicates neither a tree edge nor another closure edge;
5. active closure triples are ordered lexicographically by endpoint pair; and
6. at most K_{cl} closure slots are active and every unused slot carries designated null endpoint and bond states.

The raw decoder returns a typed failure on every malformed state:

$$\text{Dec} : \bar{\mathfrak{X}}_{N, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (40)$$

C.2 SPARSE GRAPH REPRESENTATION

Lemma C.1 (Tree residual count). *Let $G = (V, E)$ be a connected finite graph and let T be any spanning tree of G . Then*

$$|E \setminus E(T)| = |E| - |V| + 1 = \beta(G). \quad (41)$$

Proof. A spanning tree on $|V|$ vertices contains exactly $|V| - 1$ edges. Because $E(T) \subseteq E$,

$$|E \setminus E(T)| = |E| - |E(T)| = |E| - (|V| - 1) = |E| - |V| + 1. \quad (42)$$

The final expression is the cyclomatic number of a connected graph. $\square$

Theorem C.2 (Correctness and bounded completeness of the sparse representation). *The decoder*

$$\text{Dec} : \mathfrak{X}_{N, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N, K_{\text{cl}}} \quad (43)$$

is well defined and surjective up to label-preserving graph isomorphism:

1. every $X \in \mathfrak{X}_{N, K_{\text{cl}}}$ decodes to a graph in $\mathfrak{G}_{N, K_{\text{cl}}}$; and
2. for every $G \in \mathfrak{G}_{N, K_{\text{cl}}}$, there exists $X \in \mathfrak{X}_{N, K_{\text{cl}}}$ such that $\text{Dec}(X) \cong G$.

Proof. Let $G = (V, E, \ell_V, \ell_E) \in \mathfrak{G}_{N, K_{\text{cl}}}$, and choose a spanning tree $T = (V, E_T)$ with $E_T \subseteq E$. Choose a root of T and enumerate its vertices $v_1, \dots, v_n$ in any root-first traversal. Every non-root vertex then appears after its parent.

Assign node i the label of v_i . For each $i > 1$, let $\pi_i < i$ be the index of the parent of v_i in T , and assign the corresponding tree-edge label. By Lemma C.1, the residual edge set $E \setminus E_T$ contains $\beta(G) \leq K_{\text{cl}}$ edges. Orient each residual edge by increasing endpoint index, attach its original edge label, and sort the resulting triples lexicographically by endpoint pair. The resulting state is an element of $\mathfrak{X}_{N, K_{\text{cl}}}$.

The decoder reconstructs every edge in E_T from the parent pointers and every edge in $E \setminus E_T$ from the closure triples. It also preserves all node and edge labels. The map $i \mapsto v_i$ is therefore a label-preserving isomorphism from $\text{Dec}(X)$ to G .

Conversely, let $X \in \mathfrak{X}_{N, K_{\text{cl}}}$ have n active nodes. Since $\pi_i < i$, repeatedly following parent pointers from any node strictly decreases its index and must terminate at the root. The $n - 1$ parent edges therefore form a connected acyclic graph, hence a spanning tree. If the state contains $k \leq K_{\text{cl}}$ closure edges, the decoded graph has

$$|E| = (n - 1) + k \quad (44)$$

distinct edges. Its cyclomatic number is consequently

$$|E| - |V| + 1 = (n - 1 + k) - n + 1 = k \leq K_{\text{cl}}. \quad (45)$$

The remaining well-formedness conditions preserve the declared node and edge vocabularies and exclude loops or duplicate edges. Thus $\text{Dec}(X) \in \mathfrak{G}_{N, K_{\text{cl}}}$. $\square$

Global completeness does not by itself imply compatibility with a program face Ω_P . FORGE therefore restricts the morphology prior to

$$\mathcal{P}_{\text{adm}} = \{P : \mathfrak{X}_{N, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\} \quad (46)$$

and requires $\text{Enc}_P(G, \alpha) \in \mathfrak{X}_{N, K_{\text{cl}}} \cap \Omega_P$ for every admitted corpus record. We checked this adapter-level property during corpus construction.

C.3 REACTION-PROGRAM SUPPORT AND CATALOGUE REACHABILITY

Reaction programs constrain how a lipid is assembled without specifying the molecular structures that occupy its precursor roles. Let $\mathcal{G}_{\text{bounded}} \subseteq \mathcal{G}$ denote the molecular graphs within the declared atom and bond vocabularies, the 194-heavy-atom limit, and the three-closure limit. For chemistry c , let $\mathcal{D}_c(G)$ be the candidate exact-decomposition search space and define

$$\mathcal{D}_P(G) = \{(C, \tau) \in \mathcal{D}_c(G) : (G, C, \tau) \models P\}. \quad (47)$$

The valid, exact-L1 molecular space associated with P is

$$\mathcal{G}_{\text{L1}}(P) = \left\{ G \in \mathcal{G}_{\text{bounded}} : \text{Valid}(G) = 1, \widehat{\text{L1}}(G, P) = 1 \right\}. \quad (48)$$

A generated lipid can pass the exact L1 check even when one or more recovered precursors never appeared in the training component set, provided the specified reaction reconstructs the lipid from them. For a reaction program P , let $\mathfrak{C}_P$ denote its admissible precursor-program inputs, including precursor identities, role assignments, and program depth. The registered forward operator maps each input to a finite set of canonical products:

$$\mathcal{F}_P : \mathfrak{C}_P \longrightarrow 2^{\mathfrak{G}_{N, K_{\text{cl}}}}. \quad (49)$$

Given role-specific catalogues $\mathcal{C}$, let $\mathfrak{C}_P(\mathcal{C}) \subseteq \mathfrak{C}_P$ contain the admissible program inputs whose components all belong to their corresponding catalogues. The catalogue-reachable product set is

$$\mathcal{G}_{\text{cat}}(P, \mathcal{C}) = \bigcup_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} \mathcal{F}_P(\mathbf{C}). \quad (50)$$

For any graph G , define its complete program-relative preimage as

$$\mathcal{E}_P^*(G) = \{\mathbf{C} \in \mathfrak{C}_P : G \in \mathcal{F}_P(\mathbf{C})\}. \quad (51)$$

Definition C.3 (Strong catalogue escape).

$$\text{Escape}^*(G; P, \mathcal{C}) = \mathbf{1}[\mathcal{E}_P^*(G) \neq \emptyset \wedge \mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C}) = \emptyset]. \quad (52)$$

Proposition C.4 bounds the catalogue-reachable product set. For a verified exact-L1 graph, direct membership in the canonical product-hash set gives the operational strong-escape test

$$\text{StrongEscape}(G; P, \mathcal{C}) = \mathbf{1}[\widehat{\text{L1}}(G, P) = 1 \wedge G \notin \mathcal{G}_{\text{cat}}(P, \mathcal{C})]. \quad (53)$$

The precursor-input projection of $\widehat{\mathcal{E}}_P(G)$ from Appendix A.5 is a subset of $\mathcal{E}_P^*(G)$ because every returned trace passes exact forward replay. Establishing strong catalogue escape requires excluding every all-catalogue decomposition through exhaustive reverse enumeration or an independent check of complete catalogue reachability. The present experiments did not precompute a complete common-reference product-hash set, so they report verifier-recovered component novelty as a secondary diagnostic and do not report StrongEscape for FORGE. FORGE parameterizes $p_\theta(G \mid P)$ over whole graphs in $\mathcal{G}_{\text{bounded}}$, without including any component catalogue $\mathcal{C}_r$ in the generative state. Reaction programs constrain assembly semantics, while the flow determines precursor identity at atom and bond resolution.

Proposition C.4 (Finite-component reachability). *Assume that $\mathfrak{C}_P(\mathcal{C})$ is finite and that every forward-outcome set is finite. Then*

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq \sum_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|. \quad (54)$$

If $\mathfrak{C}_P(\mathcal{C}) \neq \emptyset$ and

$$m_{\text{max}, P}(\mathcal{C}) = \max_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|, \quad (55)$$

then

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq m_{\text{max}, P}(\mathcal{C}) |\mathfrak{C}_P(\mathcal{C})|. \quad (56)$$

Moreover,

$$G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C}) \iff \mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C}) \neq \emptyset. \quad (57)$$

Proof. The reachable set is a finite union of forward-outcome sets. The cardinality of a union is at most the sum of the cardinalities of its members, which proves the first inequality. Bounding each summand by $m_{\text{max}, P}(\mathcal{C})$ proves the second.

By definition, G belongs to the union if and only if at least one $\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})$ satisfies $G \in \mathcal{F}_P(\mathbf{C})$. This is equivalent to membership in $\mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C})$. $\square$

Corollary C.5 (Catalogue assemblers cannot strongly escape). *For every $G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C})$, $\text{Escape}^*(G; P, \mathcal{C}) = 0$. Consequently any assembler whose outputs are contained in $\mathcal{G}_{\text{cat}}(P, \mathcal{C})$ has zero strong-catalogue-escape yield per attempt.*

Proof. If $G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C})$, a catalogue tuple belongs to $\mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C})$, contradicting the second conjunct in the definition of strong escape. $\square$

C.4 PROGRAM-MASKED DISCRETE FLOW

Conditional path. For one coordinate with alphabet $\mathcal{X}_i$, full-support source marginal $q_{0,i}$ and clean endpoint $a \in \mathcal{X}_i$, the linear conditional path is

$$q_{t|1,i}(u | a) = (1 - t)q_{0,i}(u) + t \mathbf{1}\{u = a\}, \quad t \in [0, 1], \quad (58)$$

with a univariate conditional rate $R_{t,i}^*(u, v | a)$ that realizes this path (Qin et al., 2025). FORGE applies this construction independently to the variable coordinates conditional on the shared clean endpoint. Consequently the joint conditional path is a product of coordinate paths, not the mixture $(1 - t)q_0 + tq_1$. Its continuous-time generator permits one coordinate jump at an infinitesimal time. The denoiser predicts the clean posterior marginal of each coordinate, which is exactly the quantity needed to marginalize the corresponding local rate. The rates are derived below; Appendix A.4 specifies the capped independent-coordinate Euler approximation used in code.

Program-compatible state space and rates. For a program P , let $\mathcal{I}(P)$ index all categorical coordinates, with coordinate alphabet $\mathcal{X}_i$. Partition it into fixed coordinates $\mathcal{F}(P)$ and variable coordinates $\mathcal{V}(P)$. Let $x_{P,i}$ be the prescribed value for $i \in \mathcal{F}(P)$ and define

$$\Omega_P = \{x : x_i = x_{P,i} \text{ for every } i \in \mathcal{F}(P)\}. \quad (59)$$

The projection Π_P restores $x_{P,i}$ on $\mathcal{F}(P)$ and is the identity on $\mathcal{V}(P)$.

The program-compatible source factorizes as

$$q_0^P(x) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{P,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} q_{0,i}^P(x_i), \quad q_{0,i}^P(a) > 0 \quad \forall a \in \mathcal{X}_i. \quad (60)$$

For a clean endpoint $x_1 \in \Omega_P$, the independent-coordinate conditional path is

$$q_{t|1}^P(x | x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{1,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1 - t)q_{0,i}^P(x_i) + t \mathbf{1}\{x_i = x_{1,i}\}]. \quad (61)$$

Thus each coordinate is linearly interpolated conditional on the endpoint, while the joint law is their product. It is not the joint mixture $(1 - t)q_0^P + tq_1^P$.

For coordinate i , write $p_{t,i}(v | a) = (1 - t)q_{0,i}^P(v) + t \mathbf{1}\{v = a\}$, $\dot{p}_{t,i}(v | a) = \mathbf{1}\{v = a\} - q_{0,i}^P(v)$ and $S_{t,i}(a) = \{v : p_{t,i}(v | a) > 0\}$. The implementation’s minimum univariate conditional rate is, for $u \neq v$,

$$R_{t,i}^*(u, v | a) = \frac{[\dot{p}_{t,i}(v | a) - \dot{p}_{t,i}(u | a)]_+ \mathbf{1}\{v \in S_{t,i}(a)\}}{|S_{t,i}(a)| p_{t,i}(u | a)}, \quad (62)$$

$$R_{t,i}^*(u, u | a) = - \sum_{v \neq u} R_{t,i}^*(u, v | a). \quad (63)$$

Full source support makes the off-diagonal denominator positive for the integration times used by the sampler. For $x \neq y$, define the local joint conditional generator

$$R_t^*(x, y | x_1, P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^*(x_i, y_i | x_{1,i}). \quad (64)$$

It permits at most one coordinate to jump at an infinitesimal time. The denoiser induces the ideal marginal coordinate rate

$$R_{t,i}^\theta(u, v | x, P) = \sum_{a \in \mathcal{X}_i} p_{\theta,i}(a | x, t, P) R_{t,i}^*(u, v | a), \quad (65)$$

and hence, off diagonal,

$$R_t^\theta(x, y | P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^\theta(x_i, y_i | x, P), \quad R_t^\theta(x, x | P) = - \sum_{y \neq x} R_t^\theta(x, y | P). \quad (66)$$

Fixed coordinates have identically zero off-diagonal rates.

For a grid $0 = t_0 < \dots < t_{L_{\text{samp}}} = 1$ and $\Delta t_k = t_{k+1} - t_k$, an uncapped independent-coordinate Euler kernel would use

$$K_{k,i}^\theta(v | x) = \Delta t_k R_{t_k,i}^\theta(x_i, v | x, P), \quad v \neq x_i, \quad (67)$$

$$K_{k,i}^\theta(x_i | x) = 1 - \Delta t_k \sum_{v \neq x_i} R_{t_k,i}^\theta(x_i, v | x, P), \quad (68)$$

and $\tilde{K}_k^\theta(y | x, P) = \mathbf{1}[y_{\mathcal{F}(P)} = x_{\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} K_{k,i}^\theta(y_i | x)$. These probabilities are nonnegative when $\Delta t_k \sum_{v \neq x_i} R_{t_k,i}^\theta(x_i, v | x, P) \leq 1$ for every coordinate and context.

The actual implementation uses one Monte Carlo clean endpoint $A_{k,i} \sim p_{\theta,i}(\cdot | X_k, t_k, P)$ per coordinate. It substitutes $R_{t_k,i}^*(\cdot, \cdot | A_{k,i})$ and multiplies every off-diagonal mass by

$$\alpha_{k,i} = \min \left\{ 1, \frac{0.999}{\Delta t_k \sum_{v \neq X_{k,i}} R_{t_k,i}^*(X_{k,i}, v | A_{k,i})} \right\}, \quad (69)$$

with $\alpha_{k,i} = 1$ when the denominator is zero. The diagonal receives the remaining mass. Coordinates are then drawn independently, $X_{k+1} \sim \prod_i \hat{K}_{k,i}$, followed by Π_P . This cap makes every finite-step kernel nonnegative and stochastic. It also means the code approximates the ideal CTMC through finite Euler steps and Monte Carlo endpoint draws. It does not perform exact Gillespie simulation.

Proposition C.6 (Program-fixed invariance). *For every $P \in \mathcal{P}_{\text{adm}}$, corruption and numerical sampling satisfy $\Pr[X_t \in \Omega_P] = 1$ and $\Pr[X_k \in \Omega_P \text{ for all } k = 0, \dots, L_{\text{samp}}] = 1$.*

Proof. The source and conditional path assign zero mass outside the fixed face. Fixed coordinates also have zero off-diagonal rates. Finally, $\Pi_P(x) \in \Omega_P$ for every raw x , so the implementation safeguard preserves the claim by induction over finite steps. $\square$

Let $X_1 = \text{Enc}_P(G, \alpha)$ and define the population law

$$\nu(dG, d\alpha, dP, dt, dX_t) = \mu_{\text{bal}}(dG, d\alpha, dP) \mathcal{U}(dt) q_{t|1}^P(dX_t | X_1). \quad (70)$$

For $i \in \mathcal{V}(P)$, write $\rho_i(a | X_t, t, P) = \Pr_\nu[X_{1,i} = a | X_t, t, P]$.

Proposition C.7 (Bayes optimality of the primary coordinatewise masked-denoising loss). *Assume an unrestricted coordinatewise conditional model class and positive state-family weight. On the event $i \in \mathcal{V}(P)$, every population minimizer of the primary masked denoising term satisfies*

$$p_{\theta,i}^*(a | X_t, t, P) = \rho_i(a | X_t, t, P) \quad (71)$$

for every $a \in \mathcal{X}_i$, ν -almost surely. This proposition does not characterize the optimizer produced by auxiliary losses, finite capacity or PCGrad.

Proof. Fix a variable coordinate and context. Its conditional cross-entropy is $H(\rho_i) + D_{\text{KL}}(\rho_i \| p_{\theta,i})$, which is minimized exactly at $p_{\theta,i} = \rho_i$. The positive family weight and positive normalization by the number of variable coordinates do not alter that pointwise minimizer. Fixed coordinates do not appear in the primary loss. $\square$

Corollary C.8 (Population recovery of the local rate). *Suppose the conditional joint generator decomposes coordinatewise as above. Substituting any population minimizer $p_{\theta,i}^*$ of the primary masked denoising term into the marginal-rate formula recovers the population local generator R_t almost surely on the training path.*

Proof. The proposition identifies $p_{\theta,i}^*$ with the true clean-endpoint posterior marginal. The displayed rate is linear in that marginal, so substituting it performs the conditional expectation of $R_{t,i}^*$ required for each local coordinate generator. Summing the one-coordinate terms recovers the joint local generator. $\square$

Theorem C.9 (Ideal population correctness of program-conditioned generation). *Fix a program P with positive mass under μ_{bal} , and let*

$$q_1^P = \mathcal{L}(\text{Enc}_P(G, \alpha) \mid P), \quad \mu_G^P = \mathcal{L}([G]_{\cong} \mid P) \quad (72)$$

be the encoded endpoint law and corresponding conditional graph law. Assume finite coordinate alphabets and a full-support variable-coordinate source; every admitted endpoint decodes by exact round trip, is molecularly valid and satisfies $\widehat{\text{L1}}(G, P) = 1$; the unrestricted coordinatewise model is at a population minimum of the primary masked denoising term; and sampling starts from the program source and exactly simulates the induced local continuous-time Markov chain. Then, writing $G_1 = \text{Dec}(X_1)$, the ideal terminal state satisfies

$$\mathcal{L}(X_1 \mid P) = q_1^P, \quad \text{Dec}_{\#} q_1^P = \mu_G^P, \quad (73)$$

$$\Pr[\text{Valid}(G_1) = 1, \widehat{\text{L1}}(G_1, P) = 1 \mid P] = 1. \quad (74)$$

Formal path and proof of Theorem C.9. For the fixed program in the theorem, write the encoded endpoint mass function and population path as

$$q_1^P(x) = \Pr_{\mu_{\text{bal}}}[\text{Enc}_P(G, \alpha) = x \mid P], \quad (75)$$

$$q_t^P(x) = \sum_{x_1} q_1^P(x_1) q_{t|1}^P(x \mid x_1). \quad (76)$$

The theorem’s exact-simulation assumption uses the displayed rates $R_{t,i}^*$ on $0 \leq t < 1$, initial law $X_0 \sim q_0^P$ and terminal law defined by $t \uparrow 1$. We prove the stronger path identity

$$\mathcal{L}(X_t \mid P) = q_t^P \quad (0 \leq t < 1), \quad \mathcal{L}(X_1 \mid P) = q_1^P. \quad (77)$$

For clarity, its conditional graph mass function is

$$\mu_G^P([G]_{\cong}) = \Pr_{\mu_{\text{bal}}}[[G']_{\cong} = [G]_{\cong} \mid P], \quad (78)$$

and the remaining conclusions are

$$\text{Dec}_{\#} q_1^P = \mu_G^P \quad (79)$$

on $\mathfrak{G}_{N, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}$, with zero mass at $\perp_{\text{dec}}$, and

$$G_1 := \text{Dec}(X_1), \quad \Pr[G_1 \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(G_1) = 1, \widehat{\text{L1}}(G_1, P) = 1 \mid P] = 1. \quad (80)$$

Proof of Theorem C.9. Fix an endpoint x_1 . For one coordinate, let $p_t(v) = p_{t,i}(v \mid x_{1,i})$ and $\dot{p}_t(v) = \dot{p}_{t,i}(v \mid x_{1,i})$. The off-diagonal definition of $R_{t,i}^*$ gives the net probability flux into state v as

$$\sum_u p_t(u) R_{t,i}^*(u, v \mid x_{1,i}) = \frac{1}{|S_{t,i}|} \sum_{u \in S_{t,i}} ([\dot{p}_t(v) - \dot{p}_t(u)]_+ - [\dot{p}_t(u) - \dot{p}_t(v)]_+) \quad (81)$$

$$= \dot{p}_t(v) - \frac{1}{|S_{t,i}|} \sum_{u \in S_{t,i}} \dot{p}_t(u) = \dot{p}_t(v), \quad (82)$$

because the derivative of a probability vector sums to zero. Thus each univariate rate realizes its coordinate path. The sum of one-coordinate generators realizes the product conditional path $q_{t|1}^P$ by the product rule:

$$\partial_t q_{t|1}^P(y \mid x_1) = \sum_x q_{t|1}^P(x \mid x_1) R_t^*(x, y \mid x_1, P). \quad (83)$$

Averaging this identity over q_1^P and applying Bayes’ rule yields

$$\partial_t q_t^P(y) = \sum_x q_t^P(x) \sum_{x_1} \Pr[X_1 = x_1 \mid X_t = x, P] R_t^*(x, y \mid x_1, P) \quad (84)$$

$$= \sum_x q_t^P(x) R_t(x, y \mid P). \quad (85)$$

By Corollary C.8, $R_t^{\theta^*} = R_t$ almost surely on the population path. On every compact interval $[0, 1 - \eta]$, $\eta > 0$, this finite-state forward equation has a unique solution with initial law q_0^P , hence the exactly simulated CTMC has marginal q_t^P . Since each conditional coordinate path converges to its endpoint point mass, $q_t^P \rightarrow q_1^P$ as $t \uparrow 1$.

Finally, deterministic decode round trip sends every encoded endpoint to its source graph. Summing q_1^P over all endpoint encodings of one graph class therefore gives μ_G^P . The assumed endpoint validity and exact replay make the final probability-one statement immediate. $\square$

Corollary C.10 (Decoder contraction and approximate failure bound). *Under the endpoint assumptions of Theorem C.9, let $\tilde{q}_1^P$ be any approximate terminal raw-state law and define*

$$\mathcal{B}_P = \{x : \text{Dec}(x) = \perp_{\text{dec}}\} \cup \left\{x : \text{Dec}(x) = G \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(G)\widehat{\text{L1}}(G, P) = 0\right\}. \quad (86)$$

Then

$$\|\text{Dec}_{\#} \tilde{q}_1^P - \mu_G^P\|_{\text{TV}} \leq \|\tilde{q}_1^P - q_1^P\|_{\text{TV}}, \quad (87)$$

and

$$\Pr_{X \sim \tilde{q}_1^P}[X \in \mathcal{B}_P] \leq \|\tilde{q}_1^P - q_1^P\|_{\text{TV}}. \quad (88)$$

Proof. Total variation cannot increase under a measurable deterministic map, which proves the first inequality. The failure event has probability zero under q_1^P by Theorem C.9; its probability under $\tilde{q}_1^P$ is therefore at most the total-variation distance between the two raw-state laws. $\square$

The theorem applies to the ideal population setting. It does not characterize the finite-capacity Transformer trained with auxiliary losses and PCGrad, does not bound the Monte Carlo endpoint approximation or capped Euler discretization, and does not imply support outside q_1^P or strong catalogue escape. Corollary C.10 identifies the single approximation quantity whose control would transfer ideal endpoint correctness to the implemented sampler.

C.5 MATCHED-BUDGET POST-HOC FILTERING

Fix a reaction program P , and define

$$\mathcal{Y} = \mathfrak{G}_{N, K_{\text{cl}}} \cup \mathfrak{F}_{\text{gen}}, \quad A_P(y) = \mathbf{1}\left[y \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(y) = 1, \widehat{\text{L1}}(y, P) = 1\right]. \quad (89)$$

Here $\mathfrak{F}_{\text{gen}}$ is the set of generation-failure outcomes, including failed decodes, for which $A_P(y) = 0$. For a proposal law Q on $\mathcal{Y}$, conditional on a fixed trained model and requested program, let

$$Z_Q(P) = \Pr_{G \sim Q}[A_P(G) = 1] = \mathbb{E}_{G \sim Q}[A_P(G)] \quad (90)$$

denote the exact-L1 acceptance probability.

Proposition C.11 (Exact law of matched-budget post-hoc filtering). *For B conditionally independent attempts $Y_1, \dots, Y_B \sim Q$, $N_B = \sum_{b=1}^B A_P(Y_b)$ satisfies*

$$N_B \sim \text{Binomial}(B, Z_Q(P)) \quad (91)$$

for every $Z_Q(P) \in [0, 1]$. If $Z_Q(P) > 0$, rejection filtering additionally produces

$$Q_{\text{post}}(g | P) = \Pr_{G \sim Q}[G = g | A_P(G) = 1] = \frac{Q(g)A_P(g)}{Z_Q(P)}. \quad (92)$$

and consequently

$$\mathbb{E}[N_B] = BZ_Q(P), \quad (93)$$

$$\text{Var}(N_B) = BZ_Q(P)(1 - Z_Q(P)), \quad (94)$$

$$\Pr[N_B = 0] = (1 - Z_Q(P))^B. \quad (95)$$

If T_m is the number of independent attempts required to obtain m accepted products, then

$$\mathbb{E}[T_m] = \frac{m}{Z_Q(P)}. \quad (96)$$

Finally, for a program-conditioned proposal Q_P and an unconditioned proposal Q_0 ,

$$\mathbb{E}[N_B^P] - \mathbb{E}[N_B^0] = B(Z_{Q_P}(P) - Z_{Q_0}(P)). \quad (97)$$

Proof. Conditional independence makes the acceptance indicators Bernoulli with common probability $Z_Q(P)$, including the degenerate case $Z_Q(P) = 0$, so their sum is binomial. When $Z_Q(P) > 0$, Bayes’ rule gives

$$\Pr[G = g \mid A_P(G) = 1] = \frac{\Pr[A_P(G) = 1 \mid G = g] \Pr[G = g]}{\Pr[A_P(G) = 1]} \quad (98)$$

$$= \frac{A_P(g)Q(g)}{Z_Q(P)}. \quad (99)$$

The waiting time T_m has the negative-binomial distribution with success probability $Z_Q(P)$, whose expectation is $m/Z_Q(P)$. Applying the binomial expectation separately to Q_P and Q_0 and subtracting proves the final identity. $\square$

The proposition compares sampling efficiency only. It does not imply that conditioning improves $Z_Q(P)$; the matched production experiment tests that empirical claim. The binomial count is the total accepted-attempt count, not the number of distinct accepted products.

C.6 RECURSIVE ROUTE CONSTRUCTION

Lemma C.12 (Termination of bounded route assessment). *Fix $\mathfrak{B}_{\text{route}}$ with finite maximum route depth $D_{\mathfrak{B}}$. Assume finitely many candidate root decompositions, finitely many alternative expansions per molecule, finitely many reactants per expansion, and termination of every evidence, proposal and forward-verification call. Then $\text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$ terminates in $\mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}$.*

Proof. We use induction on the remaining depth $d = D_{\mathfrak{B}} - k$ at a node of depth k . When $d = 0$, the assessor performs the terminating evidence query and either accepts a terminal node, returns its typed failure, or rejects an expansion with a depth-exhaustion state. Suppose the claim holds whenever fewer than $d > 0$ levels remain. At the current node, the finite sequence of terminating proposal, evidence and verification calls returns either a typed non-expansion outcome or one of finitely many finite reactant tuples. In the latter case, each child is assessed at depth $k + 1$ and therefore has $d - 1$ levels remaining. Every child assessment terminates by the induction hypothesis, and a finite tuple of finite child trees forms a finite parent tree. The finite root-decomposition loop also terminates. Rejecting an ancestor identity additionally prevents a circular molecular dependency from being accepted, but finite depth alone establishes termination. $\square$

Implementation invariant C.13 (Fail-closed synthesis-dossier implementation). *For the frozen assessor implementation, a successful return $\text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) = T \in \mathfrak{D}$ is admitted only after checking $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$. Every other branch returns a member of the typed failure set $\mathfrak{F}_{\mathfrak{B}}$. This is an implementation invariant, not a completeness theorem for the chemical search space.*

The invariant records four independently checked conditions: the dossier is a well-formed tree rooted at G ; its selected root trace belongs to $\hat{\mathcal{E}}_P(G)$; every upstream reaction is evidence-admitted and satisfies $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$; and every leaf closes under the dated availability policy. A typed failure means only that no complete dossier was returned under the frozen contract, not that no route exists under a larger search budget, a different admitted reaction set, or a later supplier snapshot.

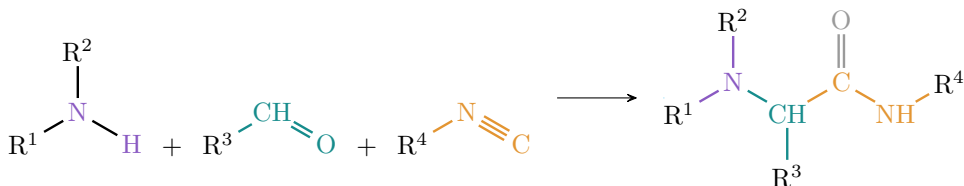
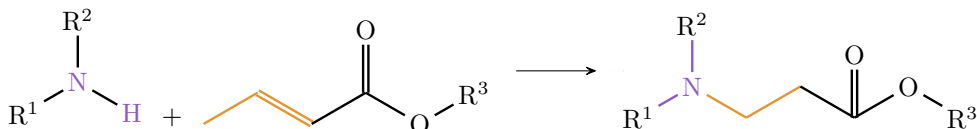
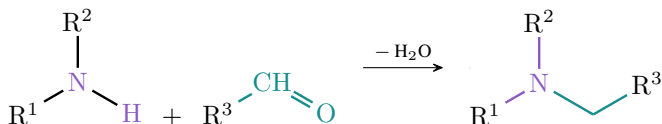
Ugi three-component reaction*amine head, aldehyde and isocyanide combine in one step; no carboxylic-acid reactant component***Repeated aza-Michael addition***the head N-H adds across the acrylate double bond; the ester is untouched***Repeated reductive amination***one new carbon-nitrogen bond; the carbonyl oxygen leaves as water*

Figure 3: **The three assembly chemistries.** Skeletal notation, with each bond carrying the colour of the component that contributes it: amine head in purple, aldehyde-derived atoms in teal, the isocyanide or acrylate tail in orange. Grey marks an atom the assembly template introduces and no precursor supplies, following the same convention as the core inset of Figure 1. Hydrogens are drawn only on the atoms whose substitution decides whether a product can be decomposed back into precursors: the head N-H, and the α -carbon C-H that the Ugi reaction takes from the aldehyde. R^i stand for the source-linked components of the frozen library, which are larger than the fragments drawn here. The Ugi variant has no carboxylic-acid reactant component.

D REACTION SCHEMES AND GENERATED EXAMPLES

The assembly schemes depict the chemistry introduced in Section 2.2. The generated examples pair molecular structures with precursor origins and exact assembly traces. Their evidentiary scope is structural and computational.

D.1 ASSEMBLY SCHEMES

The three registered assembly patterns use colour to mark precursor origin and explicit hydrogens to mark substitution constraints at the reaction core (Figure 3). Precursor-origin boundaries can differ from the functional headgroup, linker and tail regions introduced in Section 2.1.

D.1.1 REACTION TRANSFORMS AND EXACT ASSEMBLY

Each assembly chemistry defines a forward transformation that maps role-assigned precursor molecules to the product graphs permitted by that reaction. The corresponding reverse transformation decomposes a complete lipid into the precursor assignments that could have produced it. To verify a decomposition, we computationally replay the reaction on the recovered precursors. The product passes the level-one (L1) assembly check only if the reconstruction matches the original lipid exactly.

D.2 GENERATED-STRUCTURE ATLAS

Table 15 reports 12 deterministic display examples from the same frozen seed-0 Ugi population used in Figure 2. We selected the product with the lowest constitutional SHA-256 rank first, then applied ECFP4 MaxMin selection among the 23 eligible, constitutionally unique products. Selection favored structural diversity without using activity, route closure, procurement evidence or human preference. Each row includes the canonical constitutional SMILES so that the displayed identity is explicit; the exact strings are also stored in the hash-pinned atlas receipt.

Table 15: **FORGE generated-structure atlas**. The 12 rows form one multipage table, indexed by display order. Each constitutional graph is followed by its canonical SMILES. The decorated three-dimensional column shows one deterministic post-hoc RDKit ETKDGv3 structure optimized with up to 400 MMFF94s iterations. Purple denotes the amine-derived head, teal the aldehyde-derived body and tail, orange the isocyanide-derived tail and gray the assembly-introduced oxygen; these assignments are recovered through exact atom-mapped forward replay. FORGE does not generate three-dimensional coordinates, and these views are not predicted bioactive conformations. Exact L1 replay and descriptor-manifold membership are structural diagnostics, not evidence of activity, route closure or synthesis success. These are deterministic display examples, not experimental candidates.

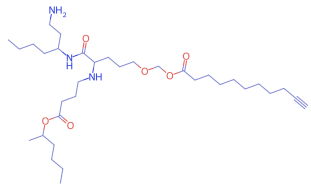
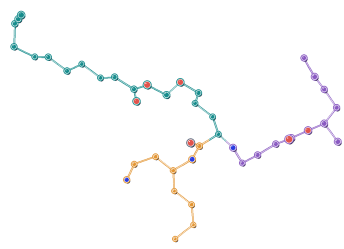
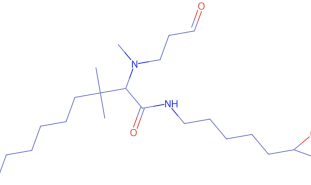
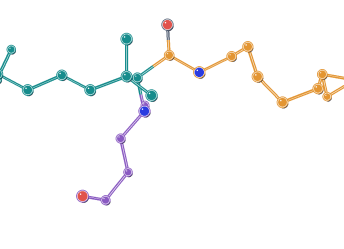
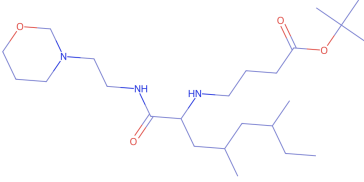
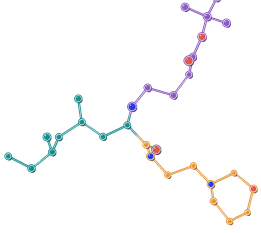
	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
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2	 <chem>CCCCC(C)(C)C(C(=O)NCCCCC1CO1)N(C)CCC=O</chem>	
3	 <chem>CCC(C)CC(C)CC(NCCCC(=O)OC(C)(C)C(=O)NCCN1CCCOC1</chem>	

Table 15, continued

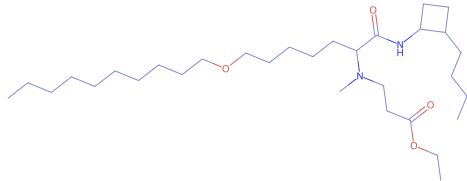
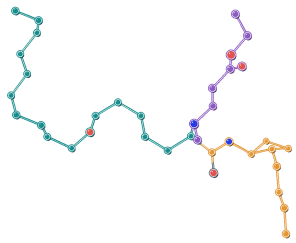
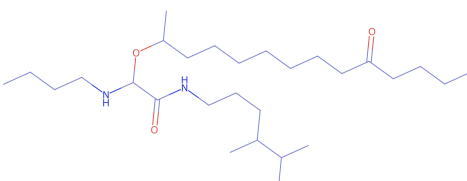
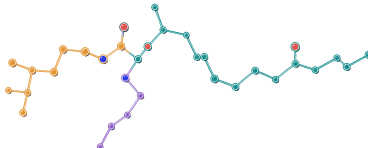
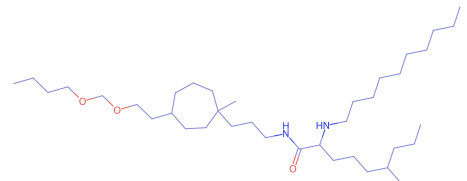
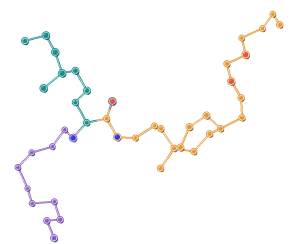
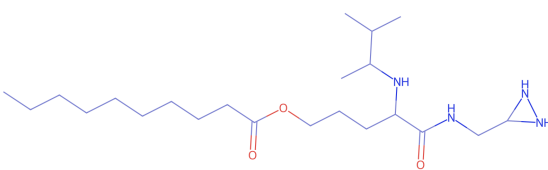
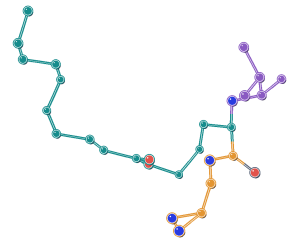
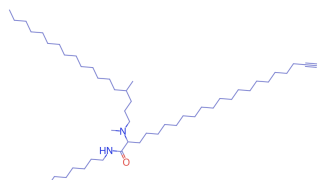
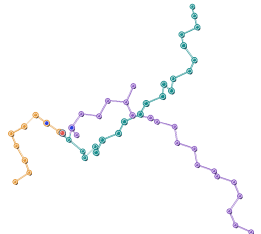
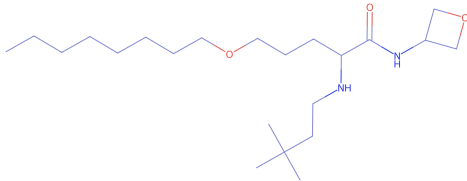
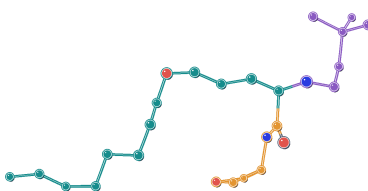
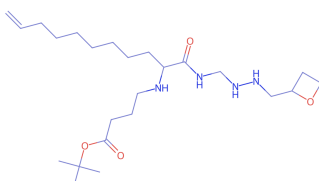
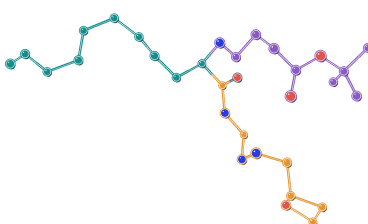
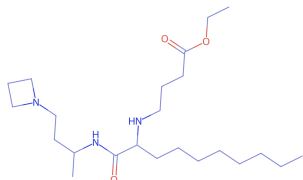
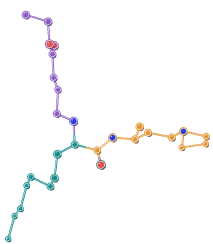
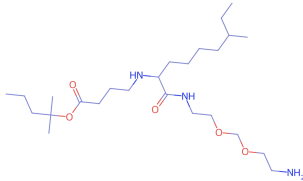
	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
4	 <chem>CCCCCCCCCOC(=O)N1CCC1CCCC)N(C)CCC(=O)OCC</chem>	
5	 <chem>CCCCNC(OC(C)CCCCCCC(=O)CCCC)C(=O)NCCCC(C)C(C)C</chem>	
6	 <chem>CCCCCCCCCNC(CCCC(C)CCC)C(=O)NCCCC1(C)CCCC(CCOC(=O)CCCC)CC1</chem>	
7	 <chem>CCCCCCCCC(=O)OCCCC(NC(C)C(C)C)C(=O)NCC1NN1</chem>	
8	 <chem>C#CCCCCCCCCCCCCCCCC(C(=O)NCCCCCCC)N(C)CCCC(C)CCCCCCCCCCCCC</chem>	

Table 15, continued

	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
9	 <chem>CCCCCCCCOCCCC(NCCC(C)(C)C)C(=O)NC1COC1</chem>	
10	 <chem>C=CCCCCCCC(NCCCC(=O)OC(C)(C)C)C(=O)NCNCCC1CCO1</chem>	
11	 <chem>CCCCCCCC(NCCCC(=O)OCC)C(=O)NC(C)CCN1CCC1</chem>	
12	 <chem>CCCC(C)(C)OC(=O)CCNC(CCCCC(C)CC)C(=O)NCCOCOCN</chem>	